

Chapter 4

3D Vector

While two-dimensional vectors are intuitive and convenient for planar geometry, real-world physical phenomena rarely confine themselves to a flat sheet of paper. Real-life motion naturally occurs in three-dimensional space, requiring three spatial parameters—length, width, and height (x, y, z) —to describe positions and displacements accurately.

Furthermore, in real-world dynamics, an object's position often changes with time. When coordinates depend on time, we obtain a **vector-valued function** (or simply a **vector function**). While a comprehensive study of vector functions lies beyond the scope of the HKDSE M2 curriculum, keeping time-dependence in mind provides a complete picture of how vectors model physical motion. In this chapter, we shall establish our foundation by focusing on constant vectors in three-dimensional space.

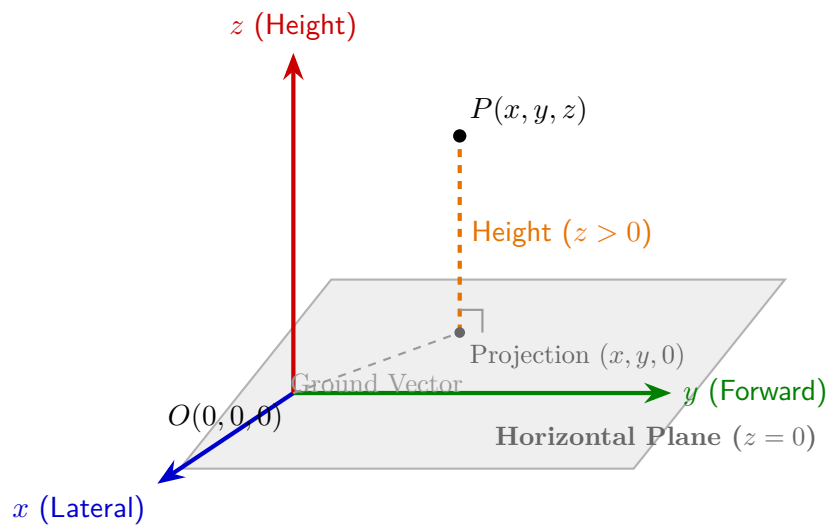


The 2D Limitation: If air traffic control only monitors the ground plane coordinates (x, y) , the airplane's trajectory appears to pass directly through the center of the mountain (x_0, y_0) , triggering a false collision warning.

The 3D Necessity: By incorporating the altitude parameter z , the 3D position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ confirms that the airplane's height z_{plane} exceeds the mountain height h at that point ($z_{\text{plane}} > h$), proving the flight path is entirely safe.

4.1 The 3D Space

To describe physical space accurately, we must extend our two-dimensional coordinate model beyond the lateral (x -axis) and forward (y -axis) directions to incorporate a vertical component representing height. Following standard mathematical convention, we denote this new vertical axis as the z -axis. In real-world applications, the xy -plane at $z = 0$ is naturally interpreted as the horizontal plane (or ground level), while points with $z > 0$ represent positions elevated above the horizon.



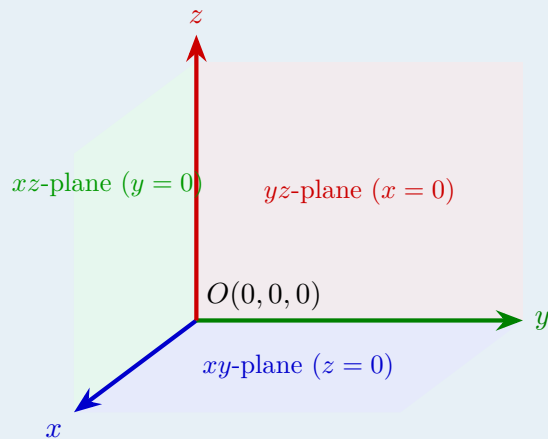
Definition 4.1.1: The 3D Space (Euclidean Space)

The **three-dimensional Euclidean space**, denoted by $\mathbb{R}^3$, is the set of all ordered triples of real numbers:

$$\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$$

Each triple (x, y, z) represents a unique point P in space relative to three mutually perpendicular axes—the x -axis, y -axis, and z -axis—intersecting at the origin $O(0, 0, 0)$. These axes divide space into three primary **coordinate planes**:

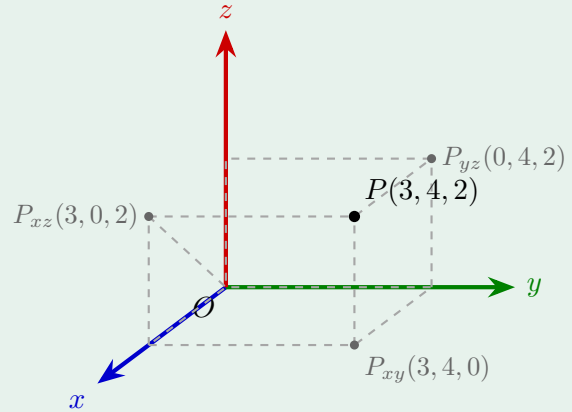
- The xy -plane (equation $z = 0$),
- The yz -plane (equation $x = 0$), and
- The xz -plane (equation $y = 0$).



Example 4.1.1: Points in 3D Space

Given a point $P(3, 4, 2)$ in 3D space.

- Find the coordinates of the perpendicular projections of P onto the xy -plane, yz -plane, and xz -plane.
- Find the coordinates of the perpendicular projections of P onto the x -axis, y -axis, and z -axis.



Solution:

- Dropping a perpendicular from $P(3, 4, 2)$ onto a coordinate plane sets the omitted coordinate to zero:
 - Projection onto the xy -plane: $P_{xy}(3, 4, 0)$
 - Projection onto the yz -plane: $P_{yz}(0, 4, 2)$
 - Projection onto the xz -plane: $P_{xz}(3, 0, 2)$
- Dropping a perpendicular onto a coordinate axis keeps only that axis's coordinate value:
 - Projection onto the x -axis: $P_x(3, 0, 0)$
 - Projection onto the y -axis: $P_y(0, 4, 0)$
 - Projection onto the z -axis: $P_z(0, 0, 2)$

Since the x -, y -, and z -axes are pairwise perpendicular (we call them **orthogonal**), the distance formula in 3D space is similar to that of 2D space, except a third coordinate component is involved.

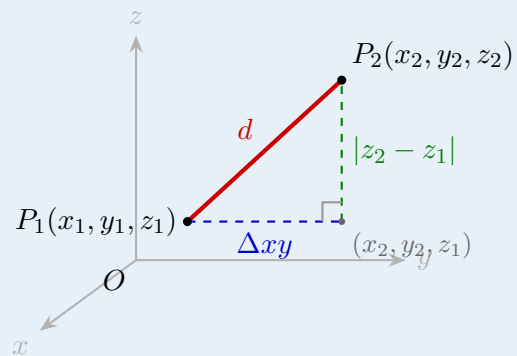
Definition 4.1.2: 3D Distance Formula

The distance d between two points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ in 3D Euclidean space is given by:

$$d = P_1P_2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

In particular, the distance from any point $P(x, y, z)$ to the origin $O(0, 0, 0)$ is:

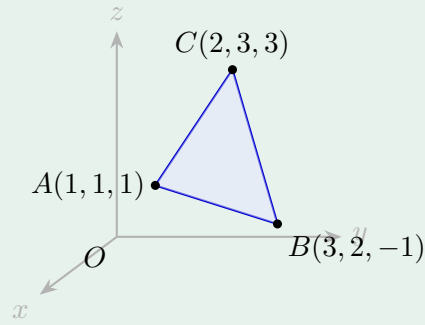
$$OP = \sqrt{x^2 + y^2 + z^2}$$



Example 4.1.2: Finding Distance Between Two Points

Given points $A(1, 1, 1)$, $B(3, 2, -1)$, and $C(2, 3, 3)$ in 3D space.

- Find the length of each side of $\triangle ABC$.
- Show that $\triangle ABC$ is an isosceles right-angled triangle.



Solution:

- Applying the 3D distance formula to each pair of vertices:

$$\begin{aligned} AB &= \sqrt{(3-1)^2 + (2-1)^2 + (-1-1)^2} = \sqrt{2^2 + 1^2 + (-2)^2} \\ &= \sqrt{4 + 1 + 4} = 3 \end{aligned}$$

$$\begin{aligned} AC &= \sqrt{(2-1)^2 + (3-1)^2 + (3-1)^2} = \sqrt{1^2 + 2^2 + 2^2} \\ &= \sqrt{1 + 4 + 4} = 3 \end{aligned}$$

$$\begin{aligned} BC &= \sqrt{(2-3)^2 + (3-2)^2 + (3-(-1))^2} = \sqrt{(-1)^2 + 1^2 + 4^2} \\ &= \sqrt{1 + 1 + 16} = \sqrt{18} = 3\sqrt{2} \end{aligned}$$

- Since $AB = AC = 3$, $\triangle ABC$ is an isosceles triangle. Furthermore, testing Pythagoras' theorem:

$$AB^2 + AC^2 = 3^2 + 3^2 = 9 + 9 = 18 = (\sqrt{18})^2 = BC^2$$

By the converse of Pythagoras' theorem, $\angle BAC = 90^\circ$. Therefore, $\triangle ABC$ is an isosceles right-angled triangle.

Exercise 4.1.1: Finding Distance Between Two Points

Given three points $P(2, -1, 3)$, $Q(4, 1, 2)$, and $R(1, 3, 7)$ in 3D space.

- Calculate the distances PQ , QR , and PR .
- Determine whether $\triangle PQR$ is an isosceles, equilateral, or scalene triangle.
- Find the coordinates of a point M on the z -axis that is equidistant from P and Q .

In two-dimensional coordinate geometry, the orientation of a line is conveniently described by a single scalar called the **gradient** (or slope). In three-dimensional space, however, a single scalar is no longer sufficient to specify direction, as a line can tilt independently in infinitely many directions relative to the three coordinate axes.

To formulate straight lines, planes, and spatial geometric relationships seamlessly in 3D space, we require a structural tool capable of encoding both magnitude and direction simultaneously—**vectors**!

Follow-up Practice**Level 1 (Basic)**

1. Given points $A(3, -1, 4)$, $B(-1, 5, 2)$, and $C(2, 0, -3)$ in 3D space.
 - (a) State the coordinates of the perpendicular projections of A onto the yz -plane and the y -axis.
 - (b) Calculate the distances AB and BC .
 - (c) Find the coordinates of the midpoint of the line segment AB .
2. A point $P(x, y, z)$ lies in the xy -plane.
 - (a) State the value of the z -coordinate of P .
 - (b) If P is equidistant from $A(1, 2, 3)$ and $B(3, -2, 1)$, show that $x - 2y = 2$.

Level 2 (Advanced)

3. The coordinates of a point K in 3D space are given by $(a, 2a, a + 1)$, where a is a real parameter.
 - (a) Express the squared distance OK^2 from the origin O to K in terms of a .
 - (b) By completing the square or otherwise, find the minimum distance from the origin O to K , and state the coordinates of K when this minimum distance occurs.
4. Points $A(2, 1, 0)$, $B(0, 3, 1)$, and $C(1, 0, 3)$ form a triangle in 3D space.
 - (a) Calculate the side lengths AB , BC , and CA .
 - (b) Show that $\triangle ABC$ is an equilateral triangle.
 - (c) Find the coordinates of a point D on the z -axis such that $DA = DB = DC$.

Level 3 (Challenge)

5. A sphere with center $S(h, k, l)$ and radius R is defined as the set of all points $P(x, y, z)$ in 3D space satisfying $SP = R$, giving the equation $(x - h)^2 + (y - k)^2 + (z - l)^2 = R^2$.
 - (a) A sphere passes through four points $O(0, 0, 0)$, $A(4, 0, 0)$, $B(0, 6, 0)$, and $C(0, 0, 8)$.
 - i. Find the coordinates of the center S and the radius R of this sphere.
 - ii. Show that for any point $P(x, y, z)$ lying on this sphere, $PA^2 + PB^2 + PC^2 + PO^2 = 4R^2$.
 - (b) A point $Q(x_0, y_0, 0)$ lies in the xy -plane inside or on $\triangle OAB$. Express the sum of squared distances $QA^2 + QB^2 + QO^2$ in terms of x_0 and y_0 , and find the coordinates of Q that minimize this sum.

4.2 3D Vector

Defining a vector in 3D space is similar to that in 2D—we are taking the directional displacement from one point to another. Let us write down the formal definition of a 3D vector.

Definition 4.2.1: 3D Vector

A **three-dimensional vector** $\vec{v}$ is represented in terms of three mutually orthogonal **elementary unit vectors** $\hat{i}$, $\hat{j}$, and $\hat{k}$ along the positive x -, y -, and z -axes respectively:

$$\vec{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k}$$

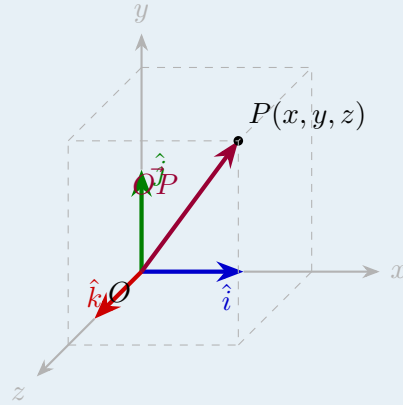
where $v_x, v_y, v_z \in \mathbb{R}$ are the scalar components of $\vec{v}$.

If $P(x, y, z)$ is a point in 3D space, its **position vector** relative to the origin $O(0, 0, 0)$ is:

$$\vec{OP} = x\hat{i} + y\hat{j} + z\hat{k}$$

The **magnitude** (or length) of $\vec{v}$ is given by:

$$|\vec{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2}$$



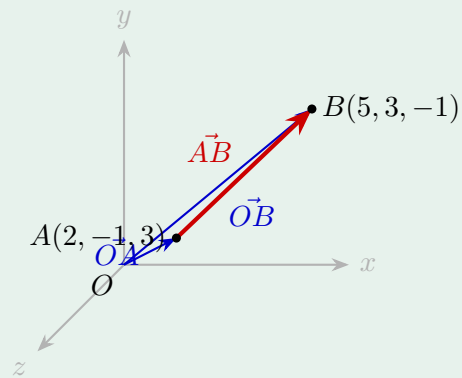
Remark

Following the convention where z is defined after x and y as the third axis, we define $\hat{k}$ after $\hat{i}$ and $\hat{j}$ as the third unit vector orthogonal to both $\hat{i}$ and $\hat{j}$.

Example 4.2.1: 3D Vector

Given two points $A(2, -1, 3)$ and $B(5, 3, -1)$ in 3D space.

- Express the position vectors $\vec{OA}$ and $\vec{OB}$ in terms of $\hat{i}$, $\hat{j}$, and $\hat{k}$.
- Find the displacement vector $\vec{AB}$ and calculate its magnitude $|\vec{AB}|$.



Solution:

- Using the coordinates of A and B :

$$\vec{OA} = 2\hat{i} - \hat{j} + 3\hat{k}, \quad \vec{OB} = 5\hat{i} + 3\hat{j} - \hat{k}$$

- The displacement vector $\vec{AB}$ is given by the triangle law of vector addition:

$$\vec{AB} = \vec{OB} - \vec{OA} = (5 - 2)\hat{i} + (3 - (-1))\hat{j} + (-1 - 3)\hat{k} = 3\hat{i} + 4\hat{j} - 4\hat{k}$$

The magnitude of $\vec{AB}$ is:

$$|\vec{AB}| = \sqrt{3^2 + 4^2 + (-4)^2} = \sqrt{9 + 16 + 16} = \sqrt{41}$$

Exercise 4.2.1: 3D Vector

Given points $P(-1, 4, 2)$ and $Q(3, 1, -2)$ relative to the origin O .

- Find the displacement vector $\vec{PQ}$ in terms of $\hat{i}$, $\hat{j}$, and $\hat{k}$.
- Calculate the magnitude $|\vec{PQ}|$.
- Find the unit vector $\hat{u}$ in the direction of $\vec{PQ}$.

The addition, subtraction, and scalar multiplication of 3D vectors follow the same component-wise algebraic rules as 2D vectors.

Proposition 4.2.1: Linearity of 3D Vectors

Let $\vec{u} = u_x\hat{i} + u_y\hat{j} + u_z\hat{k}$ and $\vec{v} = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$ be vectors in 3D space, and let $c \in \mathbb{R}$ be a scalar. Then:

- Vector Addition: $\vec{u} + \vec{v} = (u_x + v_x)\hat{i} + (u_y + v_y)\hat{j} + (u_z + v_z)\hat{k}$
- Vector Subtraction: $\vec{u} - \vec{v} = (u_x - v_x)\hat{i} + (u_y - v_y)\hat{j} + (u_z - v_z)\hat{k}$
- Scalar Multiplication: $c\vec{u} = (cu_x)\hat{i} + (cu_y)\hat{j} + (cu_z)\hat{k}$

Proof. By applying the commutative, associative, and distributive properties of vector addition and scalar multiplication over the standard orthonormal basis $\{\hat{i}, \hat{j}, \hat{k}\}$:

- For vector addition:

$$\begin{aligned}\vec{u} + \vec{v} &= (u_x\hat{i} + u_y\hat{j} + u_z\hat{k}) + (v_x\hat{i} + v_y\hat{j} + v_z\hat{k}) \\ &= (u_x\hat{i} + v_x\hat{i}) + (u_y\hat{j} + v_y\hat{j}) + (u_z\hat{k} + v_z\hat{k}) \\ &= (u_x + v_x)\hat{i} + (u_y + v_y)\hat{j} + (u_z + v_z)\hat{k}\end{aligned}$$

- For vector subtraction:

$$\begin{aligned}\vec{u} - \vec{v} &= (u_x\hat{i} + u_y\hat{j} + u_z\hat{k}) - (v_x\hat{i} + v_y\hat{j} + v_z\hat{k}) \\ &= (u_x\hat{i} - v_x\hat{i}) + (u_y\hat{j} - v_y\hat{j}) + (u_z\hat{k} - v_z\hat{k}) \\ &= (u_x - v_x)\hat{i} + (u_y - v_y)\hat{j} + (u_z - v_z)\hat{k}\end{aligned}$$

- For scalar multiplication:

$$\begin{aligned}c\vec{u} &= c(u_x\hat{i} + u_y\hat{j} + u_z\hat{k}) \\ &= c(u_x\hat{i}) + c(u_y\hat{j}) + c(u_z\hat{k}) \\ &= (cu_x)\hat{i} + (cu_y)\hat{j} + (cu_z)\hat{k}\end{aligned}$$

□

Example 4.2.2: Linear Arithmetic of 3D Vectors

Given vectors $\vec{a} = 2\hat{i} - \hat{j} + 4\hat{k}$ and $\vec{b} = 3\hat{i} + 2\hat{j} - \hat{k}$.

- Express $2\vec{a} - 3\vec{b}$ in component form.
- Calculate $|2\vec{a} - 3\vec{b}|$.

Solution:

(a) Performing componentwise scalar multiplication and subtraction:

$$\begin{aligned} 2\vec{a} - 3\vec{b} &= 2(2\hat{i} - \hat{j} + 4\hat{k}) - 3(3\hat{i} + 2\hat{j} - \hat{k}) \\ &= (4\hat{i} - 2\hat{j} + 8\hat{k}) - (9\hat{i} + 6\hat{j} - 3\hat{k}) \\ &= (4 - 9)\hat{i} + (-2 - 6)\hat{j} + (8 - (-3))\hat{k} \\ &= -5\hat{i} - 8\hat{j} + 11\hat{k} \end{aligned}$$

(b) Evaluating the magnitude:

$$|2\vec{a} - 3\vec{b}| = \sqrt{(-5)^2 + (-8)^2 + 11^2} = \sqrt{25 + 64 + 121} = \sqrt{210}$$

Exercise 4.2.2: Linear Arithmetic of 3D Vectors

Given vectors $\vec{u} = \hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{v} = 3\hat{i} - \hat{j} + 2\hat{k}$, and $\vec{w} = -2\hat{i} + 4\hat{j} + \hat{k}$.

- Find $\vec{u} + 2\vec{v} - \vec{w}$ in terms of $\hat{i}$, $\hat{j}$, and $\hat{k}$.
- Find scalars r and s such that $r\vec{u} + s\vec{v} = 7\hat{i} + 0\hat{j} + 1\hat{k}$, or show that no such scalars exist.

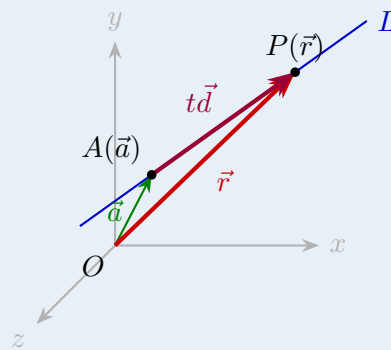
As we have developed 3D vector notation, let us revisit our problem from the previous section. To define a straight line in 3D space, we seek a representation of all points spanned along a specific direction, either forward or backward, starting from a known point.

Definition 4.2.2: Straight Line Defined by Vectors in 3D Space

A straight line L in 3D space passing through a fixed point A with position vector $\vec{a} = x_0\hat{i} + y_0\hat{j} + z_0\hat{k}$ and parallel to a non-zero **direction vector** $\vec{d} = l\hat{i} + m\hat{j} + n\hat{k}$ is defined by the **vector equation of a line**:

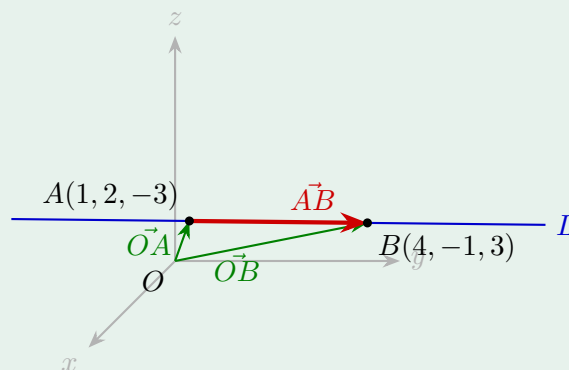
$$\vec{r}(t) = \vec{a} + t\vec{d}, \quad t \in \mathbb{R}$$

where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is the position vector of an arbitrary point $P(x, y, z)$ on L , and t is a real parameter.



Example 4.2.3: Representing a Straight Line in 3D Space Using Vectors

Find the vector equation of the straight line L passing through two given points $A(1, 2, -3)$ and $B(4, -1, 3)$.


Solution:

1. Determine the position vector of point A :

$$\vec{OA} = \hat{i} + 2\hat{j} - 3\hat{k}$$

2. Find a direction vector $\vec{d}$ by evaluating the displacement vector $\vec{AB}$:

$$\vec{d} = \vec{AB} = \vec{OB} - \vec{OA} = (4 - 1)\hat{i} + (-1 - 2)\hat{j} + (3 - (-3))\hat{k} = 3\hat{i} - 3\hat{j} + 6\hat{k}$$

3. The vector equation of line L is:

$$\vec{r}(t) = (\hat{i} + 2\hat{j} - 3\hat{k}) + t(3\hat{i} - 3\hat{j} + 6\hat{k}), \quad t \in \mathbb{R}$$

Alternatively, simplifying the direction vector to $\vec{d}' = \frac{1}{3}\vec{AB} = \hat{i} - \hat{j} + 2\hat{k}$:

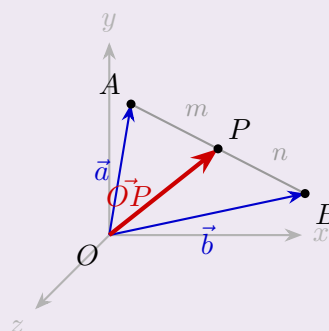
$$\vec{r}(t) = (\hat{i} + 2\hat{j} - 3\hat{k}) + t(\hat{i} - \hat{j} + 2\hat{k}), \quad t \in \mathbb{R}$$

Recall the division of a line segment in 2D geometry; we have an analogous result in 3D known as the **Section Formula** (or **Ratio Theorem**).

Theorem 4.2.1: Section Formula for 3D Vectors (Ratio Theorem)

Let A and B be two points in 3D space with position vectors $\vec{OA} = \vec{a}$ and $\vec{OB} = \vec{b}$ relative to origin O . If a point P divides the line segment AB internally in the ratio $m : n$ (such that $AP : PB = m : n$), then the position vector $\vec{OP}$ of point P is:

$$\vec{OP} = \frac{n\vec{a} + m\vec{b}}{m + n}$$



Proof. Since P lies on the segment AB such that $AP : PB = m : n$, we have $\vec{AP} = \frac{m}{m+n}\vec{AB}$.

Expressing $\vec{AB}$ in terms of position vectors $\vec{b} - \vec{a}$:

$$\vec{AP} = \frac{m}{m+n}(\vec{b} - \vec{a})$$

By the triangle law of vector addition:

$$\vec{OP} = \vec{OA} + \vec{AP} = \vec{a} + \frac{m}{m+n}(\vec{b} - \vec{a}) = \frac{(m+n)\vec{a} + m\vec{b} - m\vec{a}}{m+n} = \frac{n\vec{a} + m\vec{b}}{m+n}$$

□

Example 4.2.4: 3D Section Formula

Relative to origin O , the position vectors of two points A and B are $\vec{OA} = 2\hat{i} - \hat{j} + 5\hat{k}$ and $\vec{OB} = 7\hat{i} + 4\hat{j}$.

- Find the position vector $\vec{OP}$ and the coordinates of point P which divides line segment AB internally in the ratio $2 : 3$.
- Find the position vector of the midpoint M of line segment AB .

Solution:

- Applying the section formula for internal division with $m = 2$ and $n = 3$:

$$\vec{OP} = \frac{3\vec{OA} + 2\vec{OB}}{2+3}$$

Substituting $\vec{OA} = 2\hat{i} - \hat{j} + 5\hat{k}$ and $\vec{OB} = 7\hat{i} + 4\hat{j}$:

$$\begin{aligned}\vec{OP} &= \frac{3(2\hat{i} - \hat{j} + 5\hat{k}) + 2(7\hat{i} + 4\hat{j})}{5} = \frac{(6\hat{i} - 3\hat{j} + 15\hat{k}) + (14\hat{i} + 8\hat{j})}{5} \\ &= \frac{20\hat{i} + 5\hat{j} + 15\hat{k}}{5} = 4\hat{i} + \hat{j} + 3\hat{k}\end{aligned}$$

Thus, the coordinates of P are $(4, 1, 3)$.

- Setting $m = 1, n = 1$ for the midpoint M :

$$\vec{OM} = \frac{\vec{OA} + \vec{OB}}{2} = \frac{(2+7)\hat{i} + (-1+4)\hat{j} + (5+0)\hat{k}}{2} = \frac{9}{2}\hat{i} + \frac{3}{2}\hat{j} + \frac{5}{2}\hat{k}$$

Exercise 4.2.3: 3D Section Formula

Given points $P(1, 3, -4)$ and $Q(6, -2, 6)$ in 3D space.

- Find the position vector $\vec{OR}$ of point R which divides segment PQ internally in the ratio $3 : 2$.
- Find the coordinates of point S such that Q is the midpoint of segment PS .

To show the compatibility of 3D vectors with Cartesian geometry, we state the parametric and symmetric equations of a 3D straight line derived directly from its vector equation.

Theorem 4.2.2: Parametric and Cartesian Forms of a Straight Line

Let a line L pass through point $A(x_0, y_0, z_0)$ with direction vector $\vec{d} = l\hat{i} + m\hat{j} + n\hat{k}$.

(a) The **parametric equations** of L are:

$$x = x_0 + lt, \quad y = y_0 + mt, \quad z = z_0 + nt \quad (t \in \mathbb{R})$$

(b) If $l, m, n \neq 0$, eliminating t yields the **symmetric (Cartesian) equations** of L :

$$\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n}$$

Proof. Expanding the vector equation $\vec{r} = \vec{a} + t\vec{d}$ in component form:

$$x\hat{i} + y\hat{j} + z\hat{k} = (x_0\hat{i} + y_0\hat{j} + z_0\hat{k}) + t(l\hat{i} + m\hat{j} + n\hat{k})$$

Equating corresponding components yields the parametric equations:

$$x = x_0 + lt, \quad y = y_0 + mt, \quad z = z_0 + nt$$

Assuming $l, m, n \neq 0$, solving each equation for parameter t :

$$t = \frac{x - x_0}{l}, \quad t = \frac{y - y_0}{m}, \quad t = \frac{z - z_0}{n}$$

Equating these expressions for t gives the symmetric Cartesian equations of the line:

$$\frac{x - x_0}{l} = \frac{y - y_0}{m} = \frac{z - z_0}{n}$$

□

Example 4.2.5: Straight Line Defined by a Vector

Find the vector equation, parametric equations, and symmetric equations of the straight line L passing through point $A(2, 3, -1)$ with direction vector $\vec{d} = 4\hat{i} - \hat{j} + 5\hat{k}$.

Solution:

1. Vector equation:

$$\vec{r} = (2\hat{i} + 3\hat{j} - \hat{k}) + t(4\hat{i} - \hat{j} + 5\hat{k}), \quad t \in \mathbb{R}$$

2. Parametric equations:

$$x = 2 + 4t, \quad y = 3 - t, \quad z = -1 + 5t \quad (t \in \mathbb{R})$$

3. Symmetric Cartesian equations:

$$\frac{x - 2}{4} = \frac{y - 3}{-1} = \frac{z + 1}{5}$$

Exercise 4.2.4: Straight Line Defined by a Vector

A straight line L passes through two points $P(1, -2, 4)$ and $Q(4, 2, 1)$.

- Find a direction vector $\vec{d}$ for line L .
- Write down the vector equation and parametric equations of L .
- Write down the symmetric Cartesian equations of L .
- Determine whether the point $R(7, 6, -2)$ lies on line L .

A line's direction vector is unique only up to a non-zero scalar multiple. To establish a standardized representative direction, we frequently use the **unit direction vector** $\hat{d} = \frac{\vec{d}}{|\vec{d}|}$.

Example 4.2.6: Finding Unit Vectors That Define a Straight Line

Given the line L in symmetric form:

$$\frac{x-3}{2} = \frac{y+1}{-2} = \frac{z-4}{1}$$

- Identify a direction vector $\vec{d}$ for L .
- Find the two unit direction vectors parallel to L .

Solution:

- The denominators of the symmetric form give the components of a direction vector:

$$\vec{d} = 2\hat{i} - 2\hat{j} + \hat{k}$$

- Calculate the magnitude of $\vec{d}$:

$$|\vec{d}| = \sqrt{2^2 + (-2)^2 + 1^2} = \sqrt{4 + 4 + 1} = \sqrt{9} = 3$$

The two unit direction vectors parallel to L (pointing in opposite directions) are:

$$\hat{d} = \pm \frac{\vec{d}}{|\vec{d}|} = \pm \left(\frac{2}{3}\hat{i} - \frac{2}{3}\hat{j} + \frac{1}{3}\hat{k} \right)$$

Exercise 4.2.5: Finding Unit Vectors That Define a Straight Line

A straight line L is given by parametric equations $x = 2 - 3t$, $y = 1 + 4t$, $z = -5 + 12t$.

- State a direction vector $\vec{d}$ for L .
- Find the unit direction vector $\hat{d}$ pointing in the direction of increasing parameter t .

Remark

In 2D coordinate geometry, $Ax + By + C = 0$ represents a straight line. In 3D space, however, a single linear equation $Ax + By + Cz + D = 0$ represents a **plane**, whereas a 3D line requires **two** independent linear equations (the intersection of two non-parallel planes) or a parametric/vector system. This highlights why vector methods are indispensable in 3D geometry.

Follow-up Practice

Level 1 (Basic)

- Given vectors $\vec{a} = 3\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + 4\hat{j} + 5\hat{k}$.
 - Express $3\vec{a} + 2\vec{b}$ in component form.
 - Calculate the magnitudes $|\vec{a}|$ and $|\vec{b}|$.
 - Find the unit vector in the direction of $\vec{a} + \vec{b}$.
- A straight line L passes through $A(3, 1, -2)$ with direction vector $\vec{d} = \hat{i} - 2\hat{j} + 2\hat{k}$.
 - Write down the vector equation and parametric equations of L .
 - Express L in symmetric Cartesian form.
 - Find the coordinates of the point on L corresponding to parameter $t = 3$.

Level 2 (Advanced)

- Two lines L_1 and L_2 in 3D space are given by:

$$L_1 : \frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-1}$$

$$L_2 : \vec{r} = (4\hat{i} + 1\hat{j} + 2\hat{k}) + s(\hat{i} - \hat{j} + 2\hat{k}), \quad s \in \mathbb{R}$$

- State direction vectors $\vec{d}_1$ and $\vec{d}_2$ for L_1 and L_2 respectively.
 - Determine whether L_1 and L_2 are parallel.
 - By setting up a system of equations, determine whether L_1 and L_2 intersect. If so, find their point of intersection.
- Points $A(1, 0, 2)$, $B(3, 2, 1)$, and $C(-1, 4, 3)$ form $\triangle ABC$.
 - Find the position vector of the midpoint M of side BC .
 - Find the vector equation of the median line AM .
 - Find the position vector of the centroid G of $\triangle ABC$.

Level 3 (Challenge)

- Let L_1 be the line passing through $A(1, 2, 3)$ with direction vector $\vec{d}_1 = \hat{i} + 2\hat{j} + 3\hat{k}$, and L_2 be the line passing through $B(2, 4, 5)$ with direction vector $\vec{d}_2 = 2\hat{i} + \hat{j} - \hat{k}$.
 - Show that L_1 and L_2 are skew lines (i.e., they are neither parallel nor intersecting).
 - Let $P(1+t, 2+2t, 3+3t)$ be a variable point on L_1 and $Q(2+2s, 4+s, 5-s)$ be a variable point on L_2 .
 - Express the squared distance PQ^2 as a quadratic function of parameters t and s .
 - By minimizing PQ^2 with respect to t and s , find the minimum distance between skew lines L_1 and L_2 .
 - Verify that when PQ reaches its minimum length, the vector $\vec{PQ}$ is perpendicular to both $\vec{d}_1$ and $\vec{d}_2$.

4.3 The Scalar Product (Dot Product) of 3D Vectors

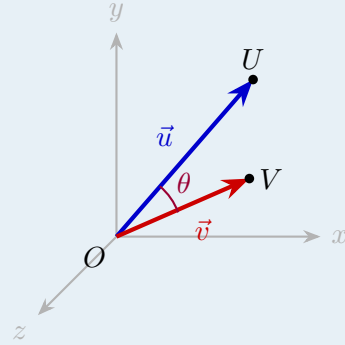
Recall the definition of the dot product in 2D geometry; we evaluate the dot product from the magnitudes of vectors and their included angle. We shall extend this concept to 3D space.

Definition 4.3.1: Scalar Product (Dot Product) of 3D Vectors

Let $\vec{u}$ and $\vec{v}$ be two non-zero vectors in 3D space, and let θ be the included angle between them such that $0 \leq \theta \leq \pi$ ($0^\circ \leq \theta \leq 180^\circ$). The **scalar product** (or **dot product**) of $\vec{u}$ and $\vec{v}$, denoted by $\vec{u} \cdot \vec{v}$, is defined as:

$$\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}| \cos \theta$$

If either $\vec{u} = \vec{0}$ or $\vec{v} = \vec{0}$, we define $\vec{u} \cdot \vec{v} = 0$.



Example 4.3.1: Finding Scalar Product with Given Included Angle

Given two vectors $\vec{u}$ and $\vec{v}$ in 3D space with magnitudes $|\vec{u}| = 5$ and $|\vec{v}| = 2\sqrt{3}$. The angle between $\vec{u}$ and $\vec{v}$ is $\theta = 30^\circ$. Calculate the scalar product $\vec{u} \cdot \vec{v}$.

Solution:

Applying the geometric definition of the 3D scalar product:

$$\vec{u} \cdot \vec{v} = |\vec{u}||\vec{v}| \cos \theta = (5)(2\sqrt{3}) \cos 30^\circ$$

Since $\cos 30^\circ = \frac{\sqrt{3}}{2}$:

$$\vec{u} \cdot \vec{v} = 10\sqrt{3} \times \frac{\sqrt{3}}{2} = 5 \times 3 = 15$$

Exercise 4.3.1: Finding Scalar Product with Given Included Angle

Given two 3D vectors $\vec{a}$ and $\vec{b}$ such that $|\vec{a}| = 4$, $|\vec{b}| = 6$, and the angle between them is 120° .

- Calculate $\vec{a} \cdot \vec{b}$.
- Determine whether the scalar product of two vectors is positive, zero, or negative when the included angle is an obtuse angle.

From this definition, the scalar products of elementary unit vectors are:

$$\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$$

and since $\hat{i}, \hat{j}, \hat{k}$ are mutually orthogonal ($\theta = 90^\circ$):

$$\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$$

Therefore, we obtain the componentwise formula for the scalar product.

Theorem 4.3.1: Scalar Product Componentwise

Let $\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$ and $\vec{b} = b_x\hat{i} + b_y\hat{j} + b_z\hat{k}$ be two vectors in 3D space. The scalar product $\vec{a} \cdot \vec{b}$ in component form is:

$$\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$$

Proof. Expanding $\vec{a} \cdot \vec{b}$ using the distributive property of dot product over vector addition:

$$\begin{aligned} \vec{a} \cdot \vec{b} &= (a_x\hat{i} + a_y\hat{j} + a_z\hat{k}) \cdot (b_x\hat{i} + b_y\hat{j} + b_z\hat{k}) \\ &= a_x b_x (\hat{i} \cdot \hat{i}) + a_x b_y (\hat{i} \cdot \hat{j}) + a_x b_z (\hat{i} \cdot \hat{k}) \\ &\quad + a_y b_x (\hat{j} \cdot \hat{i}) + a_y b_y (\hat{j} \cdot \hat{j}) + a_y b_z (\hat{j} \cdot \hat{k}) \\ &\quad + a_z b_x (\hat{k} \cdot \hat{i}) + a_z b_y (\hat{k} \cdot \hat{j}) + a_z b_z (\hat{k} \cdot \hat{k}) \end{aligned}$$

Substituting $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$ and all cross terms equal to 0:

$$\vec{a} \cdot \vec{b} = a_x b_x (1) + a_y b_y (1) + a_z b_z (1) = a_x b_x + a_y b_y + a_z b_z$$

□

Example 4.3.2: Finding Scalar Product with Given Components

Given vectors $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = 5\hat{i} + 2\hat{j} - \hat{k}$.

- Calculate $\vec{a} \cdot \vec{b}$.
- Determine whether $\vec{a}$ and $\vec{b}$ are perpendicular.

Solution:

- Applying the componentwise scalar product formula:

$$\vec{a} \cdot \vec{b} = (2)(5) + (-3)(2) + (4)(-1) = 10 - 6 - 4 = 0$$

- Since $\vec{a} \neq \vec{0}$, $\vec{b} \neq \vec{0}$, and $\vec{a} \cdot \vec{b} = 0$, the angle θ between them satisfies $\cos \theta = 0$, which implies $\theta = 90^\circ$. Thus, $\vec{a}$ and $\vec{b}$ are perpendicular.

Exercise 4.3.2: Finding Scalar Product with Given Components

Given 3D vectors $\vec{u} = 3\hat{i} + 4\hat{j} - 2\hat{k}$ and $\vec{v} = -\hat{i} + 2\hat{j} + 5\hat{k}$.

- Calculate $\vec{u} \cdot \vec{v}$.
- Calculate $\vec{u} \cdot \vec{u}$ and verify that $\vec{u} \cdot \vec{u} = |\vec{u}|^2$.

Example 4.3.3: Finding Unknowns Through Scalar Product

Two vectors are given by $\vec{a} = k\hat{i} + 2\hat{j} - 3\hat{k}$ and $\vec{b} = (k-1)\hat{i} + k\hat{j} + 2\hat{k}$, where k is a real constant.

- Express $\vec{a} \cdot \vec{b}$ in terms of k .
- If $\vec{a}$ and $\vec{b}$ are perpendicular, find all possible values of k .

Solution:

(a) Using the componentwise scalar product:

$$\begin{aligned}\vec{a} \cdot \vec{b} &= (k)(k-1) + (2)(k) + (-3)(2) \\ &= k^2 - k + 2k - 6 = k^2 + k - 6\end{aligned}$$

(b) Since $\vec{a}$ and $\vec{b}$ are perpendicular, their scalar product must be zero:

$$k^2 + k - 6 = 0 \implies (k+3)(k-2) = 0$$

Solving for k gives $k = 2$ or $k = -3$.

Exercise 4.3.3: Finding Unknowns Through Scalar Product

Given vectors $\vec{p} = 2\hat{i} + t\hat{j} + 3\hat{k}$ and $\vec{q} = t\hat{i} - 4\hat{j} + 2\hat{k}$.

- (a) If $\vec{p} \cdot \vec{q} = 8$, find the value of t .
- (b) If $\vec{p}$ is perpendicular to $\vec{q}$, find the value of t .

As in 2D vector geometry, we can determine the included angle between two 3D vectors using the cosine formula.

Theorem 4.3.2: Cosine Formula for 3D Scalar Product

Let θ be the angle between two non-zero 3D vectors $\vec{a} = a_x\hat{i} + a_y\hat{j} + a_z\hat{k}$ and $\vec{b} = b_x\hat{i} + b_y\hat{j} + b_z\hat{k}$. Then:

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{a_x b_x + a_y b_y + a_z b_z}{\sqrt{a_x^2 + a_y^2 + a_z^2} \sqrt{b_x^2 + b_y^2 + b_z^2}}$$

Proof. From Definition 3.3.1, $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$. Rearranging for $\cos \theta$:

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$$

Substituting $\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$ and magnitudes $|\vec{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$, $|\vec{b}| = \sqrt{b_x^2 + b_y^2 + b_z^2}$ yields the component formula. $\square$

Example 4.3.4: Cosine Formula in 3D Scalar Product

Find the exact included angle θ between vectors $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$.

Solution:

1. Calculate the scalar product $\vec{a} \cdot \vec{b}$:

$$\vec{a} \cdot \vec{b} = (1)(-1) + (1)(2) + (2)(1) = -1 + 2 + 2 = 3$$

2. Calculate the magnitudes $|\vec{a}|$ and $|\vec{b}|$:

$$|\vec{a}| = \sqrt{1^2 + 1^2 + 2^2} = \sqrt{1 + 1 + 4} = \sqrt{6}$$

$$|\vec{b}| = \sqrt{(-1)^2 + 2^2 + 1^2} = \sqrt{1 + 4 + 1} = \sqrt{6}$$

3. Apply the cosine formula:

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{3}{\sqrt{6}\sqrt{6}} = \frac{3}{6} = \frac{1}{2}$$

Since $0^\circ \leq \theta \leq 180^\circ$, we have $\theta = 60^\circ$.

Exercise 4.3.4: Cosine Formula in 3D Scalar Product

Given three points $A(1, 2, 0)$, $B(3, 3, 1)$, and $C(2, 0, 2)$ in 3D space.

- Find the displacement vectors $\vec{BA}$ and $\vec{BC}$ in component form.
- Calculate $\angle ABC$ correct to 1 decimal place.

Further algebraic computations can be performed when linear combinations of vectors are provided.

Example 4.3.5: Given Linear Combinations of Vectors

Let $\vec{a}$ and $\vec{b}$ be two 3D vectors such that $|\vec{a}| = 2$, $|\vec{b}| = 3$, and $|\vec{a} + \vec{b}| = \sqrt{19}$.

- Find the scalar product $\vec{a} \cdot \vec{b}$.
- Find the angle θ between $\vec{a}$ and $\vec{b}$.
- Calculate the magnitude $|2\vec{a} - 3\vec{b}|$.

Solution:

- Expanding $|\vec{a} + \vec{b}|^2$ using vector scalar product properties:

$$\begin{aligned} |\vec{a} + \vec{b}|^2 &= (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) \\ &= \vec{a} \cdot \vec{a} + 2\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{b} \\ &= |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 \end{aligned}$$

Substituting the given values:

$$\begin{aligned} (\sqrt{19})^2 &= 2^2 + 2\vec{a} \cdot \vec{b} + 3^2 \implies 19 = 4 + 2\vec{a} \cdot \vec{b} + 9 \\ 2\vec{a} \cdot \vec{b} &= 6 \implies \vec{a} \cdot \vec{b} = 3 \end{aligned}$$

- Applying the cosine formula:

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{3}{(2)(3)} = \frac{1}{2} \implies \theta = 60^\circ$$

- Evaluating $|2\vec{a} - 3\vec{b}|^2$:

$$\begin{aligned} |2\vec{a} - 3\vec{b}|^2 &= (2\vec{a} - 3\vec{b}) \cdot (2\vec{a} - 3\vec{b}) \\ &= 4|\vec{a}|^2 - 12\vec{a} \cdot \vec{b} + 9|\vec{b}|^2 \\ &= 4(2^2) - 12(3) + 9(3^2) \\ &= 16 - 36 + 81 = 61 \end{aligned}$$

Taking the square root: $|2\vec{a} - 3\vec{b}| = \sqrt{61}$.

Exercise 4.3.5: Given Linear Combinations of Vectors

Given unit vectors $\hat{u}$ and $\hat{v}$ in 3D space with an included angle of 45° .

- Evaluate $\hat{u} \cdot \hat{v}$.
- Calculate the magnitude $|3\hat{u} + \sqrt{2}\hat{v}|$.

The most important geometric application of the scalar product—finding projections—extends directly from 2D to 3D space.

Theorem 4.3.3: Projection Formulas

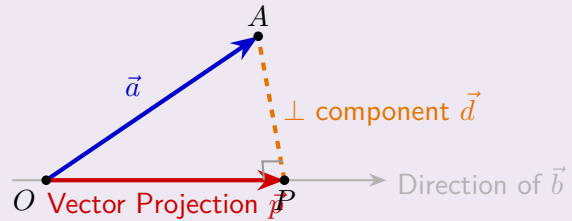
Let $\vec{a}$ and $\vec{b}$ ($\vec{b} \neq \vec{0}$) be vectors in 3D space.

- The **scalar projection** (or resolved component) of $\vec{a}$ onto $\vec{b}$ is:

$$\text{comp}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

- The **vector projection** of $\vec{a}$ onto $\vec{b}$, denoted by $\vec{p} = \text{proj}_{\vec{b}} \vec{a}$, is:

$$\vec{p} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b}$$



Proof. The vector projection $\vec{p}$ lies along $\vec{b}$, so $\vec{p} = k\vec{b}$ for some scalar k . The remaining displacement vector $\vec{d} = \vec{a} - \vec{p} = \vec{a} - k\vec{b}$ must be perpendicular to $\vec{b}$. Therefore:

$$\vec{d} \cdot \vec{b} = 0 \implies (\vec{a} - k\vec{b}) \cdot \vec{b} = 0 \implies \vec{a} \cdot \vec{b} - k(\vec{b} \cdot \vec{b}) = 0$$

Solving for k : $k = \frac{\vec{a} \cdot \vec{b}}{\vec{b} \cdot \vec{b}} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2}$. Substituting k back yields $\vec{p} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b}$. □

Example 4.3.6: Finding Projection on Another Vector

Given $\vec{a} = 4\hat{i} - 2\hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 2\hat{k}$ in 3D space.

- Find the scalar projection of $\vec{a}$ onto $\vec{b}$.
- Find the vector projection $\vec{p}$ of $\vec{a}$ onto $\vec{b}$.
- Decompose $\vec{a}$ into two orthogonal components $\vec{p}$ and $\vec{d}$ such that $\vec{p}$ is parallel to $\vec{b}$ and $\vec{d}$ is perpendicular to $\vec{b}$.

Solution:

- First evaluate $\vec{a} \cdot \vec{b}$ and $|\vec{b}|$:

$$\vec{a} \cdot \vec{b} = (4)(1) + (-2)(2) + (4)(2) = 4 - 4 + 8 = 8$$

$$|\vec{b}| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{1 + 4 + 4} = 3$$

The scalar projection is:

$$\text{comp}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{8}{3}$$

(b) The vector projection is:

$$\vec{p} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b} = \frac{8}{3^2}(\hat{i} + 2\hat{j} + 2\hat{k}) = \frac{8}{9}\hat{i} + \frac{16}{9}\hat{j} + \frac{16}{9}\hat{k}$$

(c) The perpendicular component vector $\vec{d}$ is:

$$\vec{d} = \vec{a} - \vec{p} = (4\hat{i} - 2\hat{j} + 4\hat{k}) - \left(\frac{8}{9}\hat{i} + \frac{16}{9}\hat{j} + \frac{16}{9}\hat{k} \right) = \frac{28}{9}\hat{i} - \frac{34}{9}\hat{j} + \frac{20}{9}\hat{k}$$

$$\text{Checking orthogonality: } \vec{d} \cdot \vec{b} = \frac{1}{9}(28(1) - 34(2) + 20(2)) = \frac{1}{9}(28 - 68 + 40) = 0.$$

Exercise 4.3.6: Finding Projection on Another Vector

Given vectors $\vec{u} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{v} = 3\hat{i} - 4\hat{j} + 12\hat{k}$.

- (a) Find the vector projection $\vec{p}$ of $\vec{u}$ onto $\vec{v}$.
- (b) Find the vector component of $\vec{u}$ that is perpendicular to $\vec{v}$.

Although the scalar product operates almost identically in 2D and 3D space, a unique geometric entity in 3D space is a **plane**. To define a plane, we rely on the orthogonality between a plane and a normal vector, which we explore in the next section.

Follow-up Practice

Level 1 (Basic)

1. Given vectors $\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} - k\hat{k}$, where k is a scalar.
 - (a) Calculate $\vec{a} \cdot \vec{b}$ in terms of k .
 - (b) If $\vec{a}$ and $\vec{b}$ are perpendicular, find k .
 - (c) If $k = 2$, calculate the cosine of the angle between $\vec{a}$ and $\vec{b}$.
2. Relative to origin $O(0, 0, 0)$, two points are given by $P(1, 2, 2)$ and $Q(3, 0, 4)$.
 - (a) Find position vectors $\vec{OP}$ and $\vec{OQ}$ and their magnitudes.
 - (b) Find the scalar projection and vector projection of $\vec{OP}$ onto $\vec{OQ}$.

Level 2 (Advanced)

3. Consider four points in 3D space: $A(2, 1, 0)$, $B(0, 3, 1)$, $C(1, 0, 3)$, and $D(3, 2, 4)$.
 - (a) Express displacement vectors $\vec{AB}$ and $\vec{CD}$ in component form.
 - (b) Calculate $\vec{AB} \cdot \vec{CD}$ and deduce whether line AB is perpendicular to line CD .
 - (c) Find the exact angle between line AB and line AC .
4. Let $\vec{u}$ and $\vec{v}$ be non-zero 3D vectors such that $|\vec{u}| = |\vec{v}|$.
 - (a) Prove algebraically that $(\vec{u} + \vec{v}) \cdot (\vec{u} - \vec{v}) = 0$.
 - (b) Give a geometric interpretation of this result in terms of a rhombus formed by $\vec{u}$ and $\vec{v}$.

Level 3 (Challenge)

5. Let L be a line passing through point $A(1, 2, 1)$ with direction vector $\vec{d} = 2\hat{i} - \hat{j} + 2\hat{k}$, and let $P(4, 1, 7)$ be a point not on line L .
 - (a) Write down the position vector of an arbitrary point $N(t)$ on line L in terms of parameter t .
 - (b) Express the displacement vector $\vec{PN}(t)$ in terms of t .
 - (c) Using the orthogonality condition $\vec{PN} \cdot \vec{d} = 0$, find parameter t_0 corresponding to the foot of the perpendicular N_0 from P to line L .
 - (d) Alternatively, obtain the foot of the perpendicular N_0 by evaluating the vector projection of $\vec{AP}$ onto $\vec{d}$.
 - (e) Calculate the perpendicular shortest distance from point P to line L .

4.4 Defining a Plane in 3D Space

A plane is two-dimensional. From our definition of 3D space using ordered triples and our definition of a straight line using a direction vector, we see that the dimension of a space corresponds to the number of independent variables that need to be specified. To define a plane, we can use two non-parallel vectors as a basis and express every point on the plane as a linear combination of these basis vectors. We say that such a plane is spanned by the basis vectors.

Definition 4.4.1: Plane Spanned by Two Vectors

Let A be a fixed point with position vector $\vec{a}$ in 3D space, and let $\vec{u}$ and $\vec{v}$ be two non-zero, non-parallel vectors. The **vector equation of a plane** Π passing through A and spanned by $\vec{u}$ and $\vec{v}$ is given by:

$$\vec{r}(s, t) = \vec{a} + s\vec{u} + t\vec{v}, \quad s, t \in \mathbb{R}$$

where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is the position vector of an arbitrary point $P(x, y, z)$ on Π , and s, t are independent real parameters.

Definition 4.4.2: Basis of a Plane

Two non-zero, non-parallel vectors $\vec{u}$ and $\vec{v}$ lying parallel to a plane Π form a **basis** for Π . Any vector $\vec{w}$ parallel to plane Π can be uniquely expressed as a linear combination of $\vec{u}$ and $\vec{v}$:

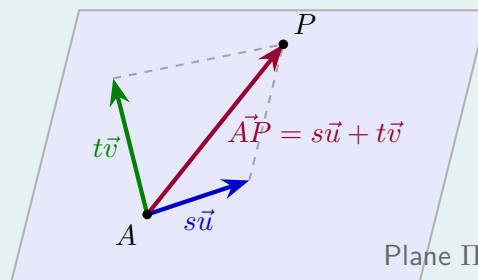
$$\vec{w} = s\vec{u} + t\vec{v} \quad \text{for unique scalars } s, t \in \mathbb{R}$$

Proposition 4.4.1: Points in a Plane as a Linear Combination of Basis Vectors

Let A be a fixed point on a plane Π with position vector $\vec{a}$, and let $\{\vec{u}, \vec{v}\}$ be a basis for Π . A point P with position vector $\vec{r}$ lies on plane Π if and only if the displacement vector $\vec{AP}$ is a linear combination of $\vec{u}$ and $\vec{v}$:

$$\vec{AP} = s\vec{u} + t\vec{v} \iff \vec{r} = \vec{a} + s\vec{u} + t\vec{v}$$

for some $s, t \in \mathbb{R}$.



Example 4.4.1: Representing Points as Linear Combinations of Basis Vectors

A plane Π passes through $A(1, 0, 2)$ with basis vectors $\vec{u} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{v} = 2\hat{i} - \hat{j} + 3\hat{k}$.

- Find the position vector and coordinates of point P corresponding to parameters $s = 2$ and $t = -1$.
- Determine whether $Q(6, 3, 3)$ lies on plane Π .

Solution:

- (a) Using the vector equation of the plane $\vec{r} = \vec{a} + s\vec{u} + t\vec{v}$:

$$\begin{aligned}\vec{OP} &= (\hat{i} + 2\hat{k}) + 2(\hat{i} + 2\hat{j} - \hat{k}) - 1(2\hat{i} - \hat{j} + 3\hat{k}) \\ &= (\hat{i} + 2\hat{k}) + (2\hat{i} + 4\hat{j} - 2\hat{k}) - (2\hat{i} - \hat{j} + 3\hat{k}) \\ &= (1 + 2 - 2)\hat{i} + (0 + 4 + 1)\hat{j} + (2 - 2 - 3)\hat{k} \\ &= \hat{i} + 5\hat{j} - 3\hat{k}\end{aligned}$$

Thus, the coordinates of P are $(1, 5, -3)$.

- (b) For $Q(6, 3, 3)$ to lie on Π , there must exist scalars $s, t \in \mathbb{R}$ satisfying $\vec{OQ} = \vec{a} + s\vec{u} + t\vec{v}$:

$$6\hat{i} + 3\hat{j} + 3\hat{k} = (\hat{i} + 2\hat{k}) + s(\hat{i} + 2\hat{j} - \hat{k}) + t(2\hat{i} - \hat{j} + 3\hat{k})$$

Equating corresponding components yields a system of three linear equations:

$$\begin{cases} 1 + s + 2t = 6 \implies s + 2t = 5 & \text{--- (1)} \\ 2s - t = 3 & \text{--- (2)} \\ 2 - s + 3t = 3 \implies -s + 3t = 1 & \text{--- (3)} \end{cases}$$

Solving (1) and (2): from (2), $t = 2s - 3$. Substituting into (1):

$$s + 2(2s - 3) = 5 \implies 5s = 11 \implies s = \frac{11}{5}, \quad t = \frac{7}{5}$$

Checking equation (3): $-\left(\frac{11}{5}\right) + 3\left(\frac{7}{5}\right) = \frac{10}{5} = 2 \neq 1$. Since the system is inconsistent, $Q(6, 3, 3)$ does not lie on plane Π .

Exercise 4.4.1: Representing Points as Linear Combinations of Basis Vectors

A plane Π is defined by $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + s(\hat{i} + \hat{j} - 2\hat{k}) + t(3\hat{i} - 2\hat{j} + \hat{k})$.

- (a) Find the position vector of the point on Π corresponding to parameters $s = 1$ and $t = 2$.
 (b) Determine whether the point $R(8, -2, -2)$ lies on plane Π .

Using this strategy, we can efficiently determine whether 3 vectors are **coplanar** or 4 points are **coplanar**.

Definition 4.4.3: Coplanar Vectors and Points

- (a) Three vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ are **coplanar** if they lie in or are parallel to the same plane. This holds if and only if one vector can be expressed as a linear combination of the other two:

$$\vec{c} = s\vec{a} + t\vec{b} \quad \text{for some } s, t \in \mathbb{R}$$

- (b) Four points A, B, C , and D are **coplanar** if they all lie on the same plane, which occurs if and only if displacement vectors $\vec{AB}$, $\vec{AC}$, and $\vec{AD}$ are coplanar.

Example 4.4.2: Showing 3 Vectors Are Coplanar

Determine whether vectors $\vec{a} = 3\hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$, and $\vec{c} = 5\hat{i} + 3\hat{j}$ are coplanar.

Solution:

We test if there exist scalars $s, t \in \mathbb{R}$ such that $\vec{c} = s\vec{a} + t\vec{b}$:

$$5\hat{i} + 3\hat{j} + 0\hat{k} = s(3\hat{i} - \hat{j} + 2\hat{k}) + t(\hat{i} + 2\hat{j} - \hat{k})$$

Equating components:

$$\begin{cases} 3s + t = 5 & \text{--- (1)} \\ -s + 2t = 3 & \text{--- (2)} \\ 2s - t = 0 & \text{--- (3)} \end{cases}$$

From (3), $t = 2s$. Substituting into (1):

$$3s + 2s = 5 \implies 5s = 5 \implies s = 1 \implies t = 2$$

Checking consistency with equation (2):

$$-s + 2t = -(1) + 2(2) = 3$$

Since $s = 1, t = 2$ satisfies all three equations, $\vec{c} = \vec{a} + 2\vec{b}$. Therefore, $\vec{a}, \vec{b}$, and $\vec{c}$ are coplanar.

Exercise 4.4.2: Showing 3 Vectors Are Coplanar

Given vectors $\vec{u} = \hat{i} + 4\hat{j} - 2\hat{k}$, $\vec{v} = 2\hat{i} - \hat{j} + 3\hat{k}$, and $\vec{w} = 4\hat{i} + 7\hat{j} - \hat{k}$. Determine whether $\vec{u}, \vec{v}$, and $\vec{w}$ are coplanar.

Example 4.4.3: Showing 4 Points Are Coplanar

Show that the four points $A(1, 2, 0)$, $B(3, 1, 2)$, $C(0, 4, -1)$, and $D(2, 6, 1)$ are coplanar.

Solution:

1. Form displacement vectors from point A :

$$\vec{AB} = (3 - 1)\hat{i} + (1 - 2)\hat{j} + (2 - 0)\hat{k} = 2\hat{i} - \hat{j} + 2\hat{k}$$

$$\vec{AC} = (0 - 1)\hat{i} + (4 - 2)\hat{j} + (-1 - 0)\hat{k} = -\hat{i} + 2\hat{j} - \hat{k}$$

$$\vec{AD} = (2 - 1)\hat{i} + (6 - 2)\hat{j} + (1 - 0)\hat{k} = \hat{i} + 4\hat{j} + \hat{k}$$

2. Set up the coplanarity equation $\vec{AD} = s\vec{AB} + t\vec{AC}$:

$$\hat{i} + 4\hat{j} + \hat{k} = s(2\hat{i} - \hat{j} + 2\hat{k}) + t(-\hat{i} + 2\hat{j} - \hat{k})$$

Equating components:

$$\begin{cases} 2s - t = 1 & \text{--- (1)} \\ -s + 2t = 4 & \text{--- (2)} \\ 2s - t = 1 & \text{--- (3)} \end{cases}$$

From (1), $t = 2s - 1$. Substituting into (2):

$$-s + 2(2s - 1) = 4 \implies 3s - 2 = 4 \implies 3s = 6 \implies s = 2, \quad t = 3$$

Checking equation (3): $2(2) - 3 = 1$, which is consistent. Since $\vec{AD} = 2\vec{AB} + 3\vec{AC}$, vectors $\vec{AB}, \vec{AC}$, and $\vec{AD}$ are coplanar, and thus points A, B, C , and D are coplanar.

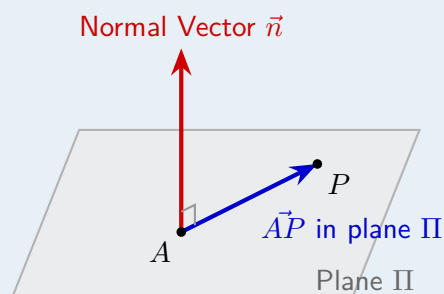
Exercise 4.4.3: Showing 4 Points Are Coplanar

Four points in 3D space are given by $P(2, -1, 1)$, $Q(4, 1, -1)$, $R(1, 2, 3)$, and $S(5, 4, k)$, where k is a real constant. Find the value of k such that points P, Q, R , and S are coplanar.

However, basis vectors of a plane are not unique—that is, you can identify a plane by any two non-parallel vectors lying on it. To establish a unique characterization, observe that the direction perpendicular to a plane is unique (up to orientation reversal). We call that perpendicular direction a **normal direction**, and the vector representing it a **normal vector**.

Definition 4.4.4: Normal Vector of a Plane

A non-zero vector $\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$ is called a **normal vector** to a plane Π if $\vec{n}$ is perpendicular (orthogonal) to every vector lying in or parallel to plane Π .



One key characteristic of a normal vector is that it is perpendicular to every vector in the plane, meaning their scalar product must be zero.

Proposition 4.4.2: Orthogonality of a Normal Vector

Let $\vec{n}$ be a normal vector to plane Π . If $A(\vec{a})$ and $P(\vec{r})$ are any two points on plane Π , then the displacement vector $\vec{AP} = \vec{r} - \vec{a}$ satisfies:

$$\vec{n} \cdot \vec{AP} = 0 \iff \vec{n} \cdot (\vec{r} - \vec{a}) = 0$$

Proof. Since A and P both lie on plane Π , displacement vector $\vec{AP} = \vec{r} - \vec{a}$ lies entirely in plane Π . By definition of a normal vector, $\vec{n}$ is orthogonal to every vector in Π . Therefore, $\vec{n} \perp \vec{AP}$, which implies $\vec{n} \cdot \vec{AP} = 0$. $\square$

From this property, a plane can be uniquely specified by a normal vector and a point on the plane.

Theorem 4.4.1: Equation of a Plane

Let Π be a plane passing through a fixed point $A(x_0, y_0, z_0)$ with position vector $\vec{a} = x_0\hat{i} + y_0\hat{j} + z_0\hat{k}$, and let $\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$ be a normal vector to Π .

(a) The **vector equation** of plane Π is:

$$\vec{n} \cdot \vec{r} = \vec{n} \cdot \vec{a} \quad \text{or} \quad \vec{n} \cdot (\vec{r} - \vec{a}) = 0$$

(b) The **Cartesian equation** (general form) of plane Π is:

$$Ax + By + Cz + D = 0$$

where $D = -(Ax_0 + By_0 + Cz_0) = -\vec{n} \cdot \vec{a}$.

Proof. Expanding $\vec{n} \cdot (\vec{r} - \vec{a}) = 0$ in component form with $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$:

$$(A\hat{i} + B\hat{j} + C\hat{k}) \cdot ((x - x_0)\hat{i} + (y - y_0)\hat{j} + (z - z_0)\hat{k}) = 0$$

By the componentwise scalar product:

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Expanding and rearranging terms:

$$Ax + By + Cz - (Ax_0 + By_0 + Cz_0) = 0$$

Setting $D = -(Ax_0 + By_0 + Cz_0)$ yields $Ax + By + Cz + D = 0$. □

Example 4.4.4: Finding Equation of a Plane from Given Normal Vector

Find the vector equation and Cartesian equation of plane Π passing through point $A(2, -1, 3)$ with normal vector $\vec{n} = 3\hat{i} + 4\hat{j} - 2\hat{k}$.

Solution:

1. Calculate the constant $\vec{n} \cdot \vec{a}$ using $\vec{a} = 2\hat{i} - \hat{j} + 3\hat{k}$:

$$\vec{n} \cdot \vec{a} = (3)(2) + (4)(-1) + (-2)(3) = 6 - 4 - 6 = -4$$

2. The vector equation of plane Π is:

$$\vec{r} \cdot (3\hat{i} + 4\hat{j} - 2\hat{k}) = -4$$

3. The Cartesian equation is obtained directly by expanding $\vec{n} \cdot \vec{r} = -4$:

$$3x + 4y - 2z = -4 \implies 3x + 4y - 2z + 4 = 0$$

Exercise 4.4.4: Finding Equation of a Plane from Given Normal Vector

Find the vector equation and Cartesian equation of the plane passing through point $P(1, 3, -2)$ with normal vector $\vec{n} = 2\hat{i} - \hat{j} + 5\hat{k}$.

Example 4.4.5: Finding Normal Vector from a Given Equation of a Plane

Given the plane $\Pi : 2x - 5y + z - 7 = 0$.

- Write down a normal vector $\vec{n}$ and the unit normal vector $\hat{n}$ to plane Π .
- Find the perpendicular distance d from origin $O(0, 0, 0)$ to plane Π .

Solution:

- The coefficients of x, y , and z in the Cartesian equation $2x - 5y + z - 7 = 0$ give the components of a normal vector:

$$\vec{n} = 2\hat{i} - 5\hat{j} + \hat{k}$$

Evaluating the magnitude of $\vec{n}$:

$$|\vec{n}| = \sqrt{2^2 + (-5)^2 + 1^2} = \sqrt{4 + 25 + 1} = \sqrt{30}$$

The unit normal vector $\hat{n}$ is:

$$\hat{n} = \frac{\vec{n}}{|\vec{n}|} = \frac{2}{\sqrt{30}}\hat{i} - \frac{5}{\sqrt{30}}\hat{j} + \frac{1}{\sqrt{30}}\hat{k}$$

- (b) The perpendicular distance d from origin $O(0,0,0)$ to plane $Ax + By + Cz + D = 0$ is given by:

$$d = \frac{|D|}{|\vec{n}|} = \frac{|-7|}{\sqrt{30}} = \frac{7}{\sqrt{30}} = \frac{7\sqrt{30}}{30}$$

Exercise 4.4.5: Finding Normal Vector from a Given Equation of a Plane

Given the plane $\Pi : 3x + 4y - 12z + 26 = 0$.

- (a) Find the unit normal vector $\hat{n}$ to plane Π .
- (b) Calculate the perpendicular distance from the origin O to plane Π .

Now we know how to categorize a plane using a normal vector. In the next section, we explore how to determine a normal vector directly from two non-parallel vectors using the **vector product (cross product)**.

Follow-up Practice

Level 1 (Basic)

- A plane Π passes through point $A(1, 2, -1)$ and has basis vectors $\vec{u} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{v} = \hat{i} + 3\hat{j} - 2\hat{k}$.
 - Write down the vector equation of plane Π in parametric form.
 - Find the coordinates of the point on Π corresponding to parameters $s = 2$ and $t = 1$.
 - Test whether point $B(4, 4, -2)$ lies on plane Π .
- Given two planes $\Pi_1 : 2x - y + 3z - 5 = 0$ and $\Pi_2 : 4x - 2y + 6z + 1 = 0$.
 - State normal vectors $\vec{n}_1$ and $\vec{n}_2$ for Π_1 and Π_2 respectively.
 - Show that $\vec{n}_2 = 2\vec{n}_1$, and deduce that planes Π_1 and Π_2 are parallel.
 - Find the perpendicular distance from the origin to each plane.

Level 2 (Advanced)

- Consider three points $A(1, 0, 1)$, $B(3, 2, -1)$, and $C(0, 4, 2)$ in 3D space.
 - Express displacement vectors $\vec{AB}$ and $\vec{AC}$ in component form.
 - Let $\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$ be a normal vector to the plane containing A , B , and C . Using $\vec{n} \cdot \vec{AB} = 0$ and $\vec{n} \cdot \vec{AC} = 0$, show that $A : B : C = 4 : 1 : 5$.
 - Deduce the Cartesian equation of the plane passing through A , B , and C .
- A variable point $P(x, y, z)$ is equidistant from two fixed points $A(1, -2, 3)$ and $B(3, 2, -1)$.
 - Using $PA^2 = PB^2$, show that the locus of P is a plane Π with equation $x + 2y - 2z - 1 = 0$.
 - Verify that the midpoint M of segment AB lies on plane Π .
 - Show that the normal vector to plane Π is parallel to displacement vector $\vec{AB}$ (plane Π is the **perpendicular bisector plane** of segment AB).

Level 3 (Challenge)

- Let Π_1 and Π_2 be two non-parallel planes defined by Cartesian equations:

$$\Pi_1 : x + 2y - z + 3 = 0, \quad \Pi_2 : 2x - y + 2z - 1 = 0$$

- State normal vectors $\vec{n}_1$ and $\vec{n}_2$ for Π_1 and Π_2 respectively, and find the cosine of the acute angle θ between the two planes.
- The two planes intersect along a line L . By setting $z = 0$ or otherwise, find a point P_0 lying on both planes.
- Let $\vec{d} = l\hat{i} + m\hat{j} + n\hat{k}$ be the direction vector of the line of intersection L . Since L lies in both planes, $\vec{d}$ must be perpendicular to both $\vec{n}_1$ and $\vec{n}_2$. Using $\vec{d} \cdot \vec{n}_1 = 0$ and $\vec{d} \cdot \vec{n}_2 = 0$, find a direction vector $\vec{d}$ for L .
- Write down the vector and symmetric Cartesian equations of the line of intersection L .

4.5 Normal Vector & Vector Product (Cross Product)

The term **vector product** (or **cross product**) produces a vector that is orthogonal to both operating vectors. We shall now examine how this operation simplifies finding normal vectors.

Proposition 4.5.1: Finding Normal Vectors From a System of Scalar Products

Let $\vec{u} = u_1\hat{i} + u_2\hat{j} + u_3\hat{k}$ and $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$ be two non-parallel vectors spanning a plane Π . A normal vector $\vec{n} = n_1\hat{i} + n_2\hat{j} + n_3\hat{k}$ orthogonal to both $\vec{u}$ and $\vec{v}$ satisfies the homogeneous linear system:

$$\begin{cases} u_1n_1 + u_2n_2 + u_3n_3 = 0 \\ v_1n_1 + v_2n_2 + v_3n_3 = 0 \end{cases}$$

Solving this system using cross-multiplication yields proportional components:

$$\frac{n_1}{u_2v_3 - u_3v_2} = \frac{n_2}{u_3v_1 - u_1v_3} = \frac{n_3}{u_1v_2 - u_2v_1}$$

Proof. Since $\vec{n}$ is orthogonal to both $\vec{u}$ and $\vec{v}$, their scalar products must be zero:

$$\vec{n} \cdot \vec{u} = 0 \implies u_1n_1 + u_2n_2 + u_3n_3 = 0 \quad \text{--- (1)}$$

$$\vec{n} \cdot \vec{v} = 0 \implies v_1n_1 + v_2n_2 + v_3n_3 = 0 \quad \text{--- (2)}$$

To eliminate n_2 , multiply equation (1) by v_2 and equation (2) by u_2 :

$$\begin{cases} u_1v_2n_1 + u_2v_2n_2 + u_3v_2n_3 = 0 \\ v_1u_2n_1 + u_2v_2n_2 + v_3u_2n_3 = 0 \end{cases}$$

Subtracting the second equation from the first:

$$(u_1v_2 - u_2v_1)n_1 + (u_3v_2 - u_2v_3)n_3 = 0 \implies \frac{n_1}{u_2v_3 - u_3v_2} = \frac{n_3}{u_1v_2 - u_2v_1} \quad \text{--- (3)}$$

Similarly, eliminating n_1 yields:

$$\frac{n_2}{u_3v_1 - u_1v_3} = \frac{n_3}{u_1v_2 - u_2v_1} \quad \text{--- (4)}$$

Combining (3) and (4) gives the proportional components:

$$\frac{n_1}{u_2v_3 - u_3v_2} = \frac{n_2}{u_3v_1 - u_1v_3} = \frac{n_3}{u_1v_2 - u_2v_1}$$

□

Remark

In my deduction, I originally considered the following system

$$\begin{cases} n_1^2 + n_2^2 + n_3^2 = |\vec{n}|^2 \\ u_1 n_1 + u_2 n_2 + u_3 n_3 = 0 \\ v_1 n_1 + v_2 n_2 + v_3 n_3 = 0 \end{cases}$$

which identifies $n_i^2 = n_i \cdot n_i$ as pseudo-constants. It is a natural way to show how determinant form works; however, not sensible for students to judge how the linearity applies. The current version provides a rational relation between components, making it clearer to deduce the normal vector as a vector parallel to a unique line.

Examining the resulting expressions reveals the bilinear form $a_1 b_2 - a_2 b_1$, which corresponds to the 2D area product. In 3D space, $\hat{i} \times \hat{j}$ evaluates to a vector orthogonal to both $\hat{i}$ and $\hat{j}$, which is $\hat{k}$.

Definition 4.5.1: The Normal Vector

Let $\vec{u} = u_1 \hat{i} + u_2 \hat{j} + u_3 \hat{k}$ and $\vec{v} = v_1 \hat{i} + v_2 \hat{j} + v_3 \hat{k}$ be two vectors in 3D space. The **normal vector** of the plane spanned by $\vec{u}$ and $\vec{v}$ is defined as:

$$\vec{w} = (u_2 v_3 - u_3 v_2) \hat{i} + (u_3 v_1 - u_1 v_3) \hat{j} + (u_1 v_2 - u_2 v_1) \hat{k}$$

which is unique up to a non-zero scalar multiple.

Example 4.5.1: Finding Normal Vectors of Two Vectors

Find a normal vector to the plane spanned by $\vec{u} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{v} = 2\hat{i} - \hat{j} + 3\hat{k}$ by setting up a system of scalar products.

Solution:

Let $\vec{n} = n_1 \hat{i} + n_2 \hat{j} + n_3 \hat{k}$ be the required normal vector. Since $\vec{n}$ must be perpendicular to both $\vec{u}$ and $\vec{v}$:

$$\begin{cases} n_1 + 2n_2 - n_3 = 0 & \text{--- (1)} \\ 2n_1 - n_2 + 3n_3 = 0 & \text{--- (2)} \end{cases}$$

Eliminating n_1 by multiplying (1) by 2 and subtracting from (2):

$$(2n_1 - n_2 + 3n_3) - 2(n_1 + 2n_2 - n_3) = 0 \implies -5n_2 + 5n_3 = 0 \implies n_2 = n_3$$

Substituting $n_2 = n_3$ into equation (1):

$$n_1 + 2n_3 - n_3 = 0 \implies n_1 = -n_3$$

Choosing $n_3 = 1$, we obtain $n_1 = -1$ and $n_2 = 1$. Thus, a normal vector is $\vec{n} = -\hat{i} + \hat{j} + \hat{k}$.

Exercise 4.5.1: Finding Normal Vectors of Two Vectors

Using a system of scalar products, find a normal vector to the plane spanned by $\vec{a} = 3\hat{i} + \hat{j} - 2\hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + \hat{k}$.

Definition 4.5.2: Equation of a Plane in 3D Space

Let Π be a plane in 3D space passing through a fixed point $A(x_0, y_0, z_0)$ with position vector $\vec{a} = x_0\hat{i} + y_0\hat{j} + z_0\hat{k}$, and let $\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$ be a non-zero normal vector orthogonal to Π .

- (a) **Vector Form:** An arbitrary point $P(x, y, z)$ with position vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ lies on plane Π if and only if:

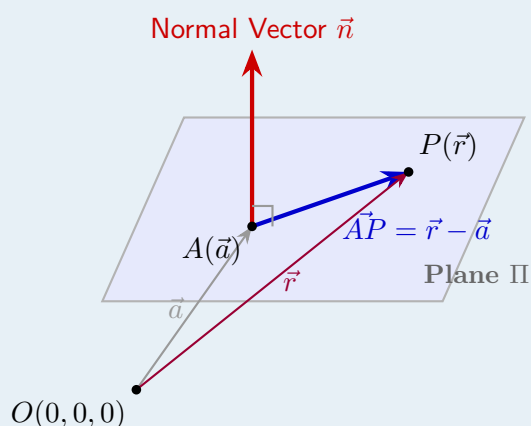
$$\vec{n} \cdot (\vec{r} - \vec{a}) = 0 \quad \text{or} \quad \vec{n} \cdot \vec{r} = d_0$$

where $d_0 = \vec{n} \cdot \vec{a} = Ax_0 + By_0 + Cz_0$ is a scalar constant.

- (b) **Cartesian Form (General Form):** Expanding the vector equation in component form yields the linear Cartesian equation:

$$Ax + By + Cz + D = 0$$

where $D = -d_0 = -(Ax_0 + By_0 + Cz_0)$. The coefficients A, B , and C directly specify the components of a normal vector $\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$ to the plane.


Example 4.5.2: Equation of a Plane from Normal Vector

Find the Cartesian equation of plane Π passing through point $A(3, -2, 1)$ with normal vector $\vec{n} = 2\hat{i} - \hat{j} + 4\hat{k}$.

Solution:

Using the point-normal equation of a plane $\vec{n} \cdot (\vec{r} - \vec{a}) = 0$:

$$2(x - 3) - 1(y - (-2)) + 4(z - 1) = 0 \implies 2x - y + 4z - 12 = 0$$

Exercise 4.5.2: Equation of a Plane from Normal Vector

Find the Cartesian equation of the plane passing through point $P(-1, 4, 2)$ with normal vector $\vec{n} = 3\hat{i} + 2\hat{j} - \hat{k}$.

Example 4.5.3: Equation of a Plane from 3 Points Through Normal Vector

Find the Cartesian equation of plane Π passing through three non-collinear points $A(1, 2, 0)$, $B(3, 0, 1)$, and $C(0, 3, 2)$.

Solution:

- Form two displacement vectors lying in plane Π :

$$\vec{AB} = 2\hat{i} - 2\hat{j} + \hat{k}, \quad \vec{AC} = -\hat{i} + \hat{j} + 2\hat{k}$$

2. Let $\vec{n} = n_1\hat{i} + n_2\hat{j} + n_3\hat{k}$ be a normal vector to Π . Since $\vec{n}$ is orthogonal to both $\vec{AB}$ and $\vec{AC}$:

$$\begin{cases} 2n_1 - 2n_2 + n_3 = 0 & \text{--- (1)} \\ -n_1 + n_2 + 2n_3 = 0 & \text{--- (2)} \end{cases}$$

Solving gives $n_3 = 0$ and $n_1 = n_2$. Setting $n_1 = 1$, we obtain $\vec{n} = \hat{i} + \hat{j}$.

3. Using point $A(1, 2, 0)$ yields:

$$1(x - 1) + 1(y - 2) + 0(z - 0) = 0 \implies x + y - 3 = 0$$

Exercise 4.5.3: Equation of a Plane from 3 Points Through Normal Vector

Given three points $P(2, 1, -1)$, $Q(4, -1, 1)$, and $R(1, 0, 2)$ in 3D space.

- Find a normal vector $\vec{n}$ to the plane passing through P, Q , and R .
- Find the Cartesian equation of the plane containing P, Q , and R .

Proposition 4.5.2: A Plane Has a Unique Normal Direction

Any two normal vectors to the same plane are non-zero scalar multiples of each other. Consequently, the normal direction of a plane is unique up to orientation reversal.

Proof. Let Π be a plane in 3D space with non-parallel basis vectors $\vec{u}$ and $\vec{v}$. A normal vector $\vec{n}$ to Π satisfies $\vec{n} \cdot \vec{u} = 0$ and $\vec{n} \cdot \vec{v} = 0$. Since the subspace orthogonal to both $\vec{u}$ and $\vec{v}$ in $\mathbb{R}^3$ is 1-dimensional, any two normal vectors $\vec{n}_1$ and $\vec{n}_2$ are parallel to $\vec{u} \times \vec{v}$, making them scalar multiples of each other ($\vec{n}_2 = k\vec{n}_1$ for $k \neq 0$). $\square$

Proposition 4.5.3: Defining a Plane by a Normal Vector

A plane is uniquely determined (up to a parallel shift) by specifying a normal vector $\vec{n}$ orthogonal to every vector in the plane.

Proof. Let $\vec{n} = A\hat{i} + B\hat{j} + C\hat{k}$ ($\vec{n} \neq \vec{0}$) be a given normal vector and $A(x_0, y_0, z_0)$ be a fixed point. A point $P(\vec{r})$ lies on plane Π if and only if $\vec{n} \cdot (\vec{r} - \vec{a}) = 0$. Expanding this gives $Ax + By + Cz + D = 0$. Any other plane orthogonal to $\vec{n}$ passing through A satisfies the identical linear equation, establishing uniqueness. $\square$

Proposition 4.5.4: The Area Product in 3D Space

Let $\vec{u} = u_1\hat{i} + u_2\hat{j} + u_3\hat{k}$ and $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$. The area product (vector product) of the parallelogram formed by $\vec{u}$ and $\vec{v}$ is:

$$\vec{u} \times \vec{v} = (u_2v_3 - u_3v_2)(\hat{j} \times \hat{k}) + (u_3v_1 - u_1v_3)(\hat{k} \times \hat{i}) + (u_1v_2 - u_2v_1)(\hat{i} \times \hat{j})$$

Proof. Expanding $\vec{u} \times \vec{v}$ using bilinearity over vector addition and setting self-cross products to zero ($\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}$), applying anti-commutativity ($\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$) to cross terms, and grouping components for coordinate planes ($\hat{j} \times \hat{k}$), ($\hat{k} \times \hat{i}$), and ($\hat{i} \times \hat{j}$) yields the result. $\square$

The magnitude of this vector product represents the area of the parallelogram:

$$|\vec{u} \times \vec{v}| = \sqrt{(u_2v_3 - u_3v_2)^2 + (u_3v_1 - u_1v_3)^2 + (u_1v_2 - u_2v_1)^2}$$

This matches the magnitude of normal vector $\vec{w}$ in Definition 5.2, establishing $\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$.

Theorem 4.5.1: Cross Product by Determinant

Let $\vec{u} = u_1\hat{i} + u_2\hat{j} + u_3\hat{k}$ and $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$ be two 3D vectors. The cross product $\vec{u} \times \vec{v}$ can be evaluated using a formal 3×3 determinant:

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} \hat{i} - \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} \hat{j} + \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \hat{k}$$

First Proof. Expanding the formal 3×3 determinant along its first row using cofactor expansion:

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \hat{i}(u_2v_3 - u_3v_2) - \hat{j}(u_1v_3 - u_3v_1) + \hat{k}(u_1v_2 - u_2v_1)$$

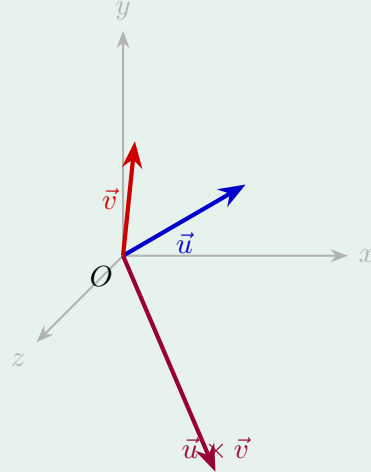
which matches Definition 5.2. □

Second Proof. Solving the homogeneous linear system $u_1n_1 + u_2n_2 + u_3n_3 = 0$ and $v_1n_1 + v_2n_2 + v_3n_3 = 0$ using Cramer's rule yields component solutions $n_1 = u_2v_3 - u_3v_2$, $n_2 = u_3v_1 - u_1v_3$, and $n_3 = u_1v_2 - u_2v_1$, reproducing the determinant expansion. □

Example 4.5.4: Finding Normal Vector by Cross Product

Given vectors $\vec{u} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{v} = \hat{i} + 3\hat{j} + 2\hat{k}$.

- Compute the normal vector orthogonal to $\vec{u}$ and $\vec{v}$ using determinant expansion.
- Find the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$.



Solution:

- Evaluating the 3×3 determinant:

$$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 3 & 2 \end{vmatrix} = 5\hat{i} - 5\hat{j} + 5\hat{k}$$

- The area of the parallelogram is:

$$\text{Area} = |\vec{u} \times \vec{v}| = \sqrt{5^2 + (-5)^2 + 5^2} = 5\sqrt{3} \text{ sq. units}$$

Exercise 4.5.4: Finding Normal Vector by Cross Product

Given $\vec{a} = 3\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{b} = -\hat{i} + 2\hat{j} + 4\hat{k}$.

- (a) Calculate the cross product $\vec{a} \times \vec{b}$.
- (b) Find the unit normal vector perpendicular to both $\vec{a}$ and $\vec{b}$.
- (c) Calculate the area of the triangle formed by $\vec{a}$ and $\vec{b}$ sharing the origin O .

Unit vectors cross-product relations follow the cyclic permutation:

$$\hat{i} \times \hat{j} = \hat{k}, \quad \hat{j} \times \hat{k} = \hat{i}, \quad \hat{k} \times \hat{i} = \hat{j}$$

Reversing the order introduces a negative sign due to anti-commutativity:

$$\hat{j} \times \hat{i} = -\hat{k}, \quad \hat{k} \times \hat{j} = -\hat{i}, \quad \hat{i} \times \hat{k} = -\hat{j}$$

Follow-up Practice

Level 1 (Basic)

1. Given vectors $\vec{u} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{v} = \hat{i} - 2\hat{j} + 3\hat{k}$.
 - (a) Calculate cross product $\vec{u} \times \vec{v}$ in component form.
 - (b) Verify that $\vec{u} \times \vec{v}$ is perpendicular to both $\vec{u}$ and $\vec{v}$ using scalar products.
 - (c) Calculate the area of the parallelogram spanned by $\vec{u}$ and $\vec{v}$.
2. Three points in 3D space are given by $A(1, 0, -1)$, $B(2, 1, 3)$, and $C(-1, 2, 1)$.
 - (a) Find displacement vectors $\vec{AB}$ and $\vec{AC}$ in component form.
 - (b) Compute $\vec{AB} \times \vec{AC}$.
 - (c) Hence, find the area of $\triangle ABC$.

Level 2 (Advanced)

3. Let $\vec{a} = 3\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$.
 - (a) Find a unit vector $\hat{n}$ perpendicular to both $\vec{a}$ and $\vec{b}$.
 - (b) Find the Cartesian equation of the plane passing through origin $O(0, 0, 0)$ and containing both $\vec{a}$ and $\vec{b}$.
 - (c) Find the Cartesian equation of a parallel plane passing through point $P(2, -1, 4)$.
4. Prove Lagrange's Identity $|\vec{a} \times \vec{b}|^2 + (\vec{a} \cdot \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2$ for any two 3D vectors $\vec{a}$ and $\vec{b}$.

Level 3 (Challenge)

5. Vertices of a tetrahedron $OABC$ are given by $O(0, 0, 0)$, $A(1, 2, 1)$, $B(3, 0, 2)$, and $C(2, 3, -1)$.
 - (a) Compute $\vec{OA} \times \vec{OB}$, and find the area of base $\triangle OAB$.
 - (b) Find the Cartesian equation of the plane containing $\triangle OAB$.
 - (c) Calculate the perpendicular height h from apex C to plane OAB .
 - (d) Evaluate the exact volume of tetrahedron $OABC$.

4.6 Projection on a Plane

To find the orthogonal projection of a vector onto a plane, we decompose the vector into two components using a normal vector to the plane.

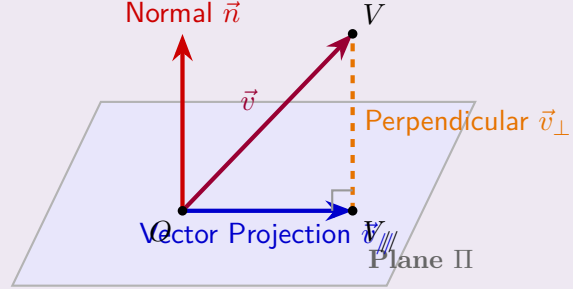
Theorem 4.6.1: Gram-Schmidt Orthogonalization for 3D Vectors

Let Π be a plane with normal vector $\vec{n} \neq \vec{0}$, and let $\vec{v}$ be any vector in 3D space. Vector $\vec{v}$ can be uniquely decomposed into two orthogonal components:

$$\vec{v} = \vec{v}_{//} + \vec{v}_{\perp}$$

where:

- (a) $\vec{v}_{\perp} = \text{proj}_{\vec{n}} \vec{v} = \left(\frac{\vec{v} \cdot \vec{n}}{|\vec{n}|^2} \right) \vec{n}$ is the component **perpendicular** to plane Π (parallel to $\vec{n}$).
- (b) $\vec{v}_{//} = \vec{v} - \vec{v}_{\perp} = \vec{v} - \left(\frac{\vec{v} \cdot \vec{n}}{|\vec{n}|^2} \right) \vec{n}$ is the **orthogonal projection** of $\vec{v}$ onto plane Π .



Proof. Since $\vec{v}_{\perp}$ is parallel to $\vec{n}$, $\vec{v}_{\perp} = k\vec{n}$ for scalar k . Component $\vec{v}_{//} = \vec{v} - k\vec{n}$ lies in plane Π , so it is orthogonal to $\vec{n}$:

$$\vec{v}_{//} \cdot \vec{n} = 0 \implies (\vec{v} - k\vec{n}) \cdot \vec{n} = 0 \implies k = \frac{\vec{v} \cdot \vec{n}}{|\vec{n}|^2}$$

Substituting k back yields $\vec{v}_{\perp} = \left(\frac{\vec{v} \cdot \vec{n}}{|\vec{n}|^2} \right) \vec{n}$ and $\vec{v}_{//} = \vec{v} - \vec{v}_{\perp}$. □

Example 4.6.1: Gram-Schmidt Process for a 3D Vector

Given vector $\vec{v} = 2\hat{i} + 3\hat{j} + 4\hat{k}$ and plane $\Pi : x + y + z = 0$.

- (a) Find a normal vector $\vec{n}$ to plane Π .
- (b) Decompose $\vec{v}$ into $\vec{v}_{//} + \vec{v}_{\perp}$, where $\vec{v}_{//}$ lies in plane Π and $\vec{v}_{\perp}$ is perpendicular to Π .
- (c) Verify that $\vec{v}_{//}$ lies in plane Π and is orthogonal to $\vec{v}_{\perp}$.

Solution:

(a) A normal vector to $x + y + z = 0$ is $\vec{n} = \hat{i} + \hat{j} + \hat{k}$.

(b) Evaluating $\vec{v} \cdot \vec{n} = 9$ and $|\vec{n}|^2 = 3$:

$$\vec{v}_{\perp} = \frac{9}{3}(\hat{i} + \hat{j} + \hat{k}) = 3\hat{i} + 3\hat{j} + 3\hat{k}$$

$$\vec{v}_{//} = (2\hat{i} + 3\hat{j} + 4\hat{k}) - (3\hat{i} + 3\hat{j} + 3\hat{k}) = -\hat{i} + \hat{k}$$

(c) Substituting components of $\vec{v}_{//} = -\hat{i} + \hat{k}$ into $x + y + z$: $(-1) + 0 + 1 = 0$, confirming it lies in Π . $\vec{v}_{//} \cdot \vec{v}_{\perp} = (-1)(3) + (0)(3) + (1)(3) = 0$, confirming orthogonality.

Exercise 4.6.1: Gram-Schmidt Process for a 3D Vector

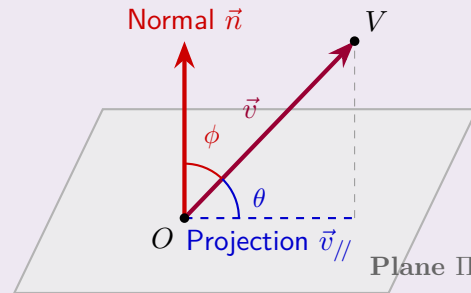
Given vector $\vec{u} = 4\hat{i} - \hat{j} + 2\hat{k}$ and plane $\Pi : 2x - y + 2z - 5 = 0$.

- Find vector projection $\vec{u}_{//}$ of $\vec{u}$ onto plane Π .
- Find perpendicular component vector $\vec{u}_{\perp}$.
- Calculate magnitude $|\vec{u}_{//}|$.

Theorem 4.6.2: Angle Between a Vector and a Plane

Let $\vec{v}$ be a non-zero vector and Π be a plane with normal vector $\vec{n}$. The acute angle θ ($0^\circ \leq \theta \leq 90^\circ$) between vector $\vec{v}$ and plane Π satisfies:

$$\sin \theta = \frac{|\vec{v} \cdot \vec{n}|}{|\vec{v}||\vec{n}|}$$



Proof. Let ϕ be the acute angle between vector $\vec{v}$ and normal vector $\vec{n}$, so $\cos \phi = \frac{|\vec{v} \cdot \vec{n}|}{|\vec{v}||\vec{n}|}$. Since $\vec{n} \perp \Pi$, angle θ between $\vec{v}$ and its projection on plane Π is complementary to ϕ ($\theta = 90^\circ - \phi$). Using $\sin \theta = \cos \phi$ gives the formula. $\square$

Example 4.6.2: Finding Angle Between a Vector and a Plane

Find the exact angle θ between vector $\vec{v} = 2\hat{i} - \hat{j} + 2\hat{k}$ and plane $\Pi : x + 2y - 2z + 5 = 0$.

Solution:

- Identify normal vector $\vec{n} = \hat{i} + 2\hat{j} - 2\hat{k}$.
- Evaluate $|\vec{v} \cdot \vec{n}| = |2 - 2 - 4| = 4$.
- Magnitudes: $|\vec{v}| = 3$, $|\vec{n}| = 3$.
- $\sin \theta = \frac{4}{9} \implies \theta = \arcsin\left(\frac{4}{9}\right) \approx 26.4^\circ$.

Exercise 4.6.2: Finding Angle Between a Vector and a Plane

A straight line L has vector equation $\vec{r} = (3\hat{i} - \hat{j} + \hat{k}) + t(\hat{i} + \hat{j} + \sqrt{2}\hat{k})$.

- Find direction vector $\vec{d}$ for line L and normal vector $\vec{n}$ to plane $\Pi : x - y + \sqrt{2}z - 3 = 0$.
- Find the exact acute angle θ between line L and plane Π .

Follow-up Practice

Level 1 (Basic)

- Given vector $\vec{a} = 3\hat{i} + 2\hat{j} - 6\hat{k}$ and plane $\Pi : 2x - y + 2z - 4 = 0$.
 - Find the scalar projection of $\vec{a}$ onto the normal direction of plane Π .
 - Find orthogonal projection vector $\vec{a}_{//}$ of $\vec{a}$ onto plane Π .
 - Calculate $\sin \theta$ for angle θ between $\vec{a}$ and plane Π .
- A straight line L passes through $A(1, 2, -1)$ and $B(3, 5, 1)$; plane Π is $2x + y - z + 3 = 0$.
 - Express direction vector $\vec{AB}$ in component form.
 - Calculate the acute angle between line L and plane Π correct to 1 decimal place.

Level 2 (Advanced)

- Consider plane $\Pi : x - 2y + 2z - 9 = 0$ and point $P(3, 4, 7)$.
 - Find parametric equations of the normal line L_N passing through P perpendicular to plane Π .
 - Find the coordinates of the foot of the perpendicular N_0 from P to plane Π .
 - Find the coordinates of reflected point P' of P across plane Π .
- Line L is $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-2}{2}$, plane Π is $x + y + 4z - 6 = 0$.
 - Find intersection point K between line L and plane Π .
 - Find vector projection of direction vector $\vec{d}$ of line L onto plane Π .
 - Write down the vector equation of projected line L' lying in plane Π .

Level 3 (Challenge)

- A light ray emitted from $S(2, 3, 5)$ along direction $\vec{d} = \hat{i} - 2\hat{j} - 2\hat{k}$ strikes mirror plane $\Pi : 2x + y - 2z + 3 = 0$ at M .
 - Find the vector equation of incident ray line L_S and coordinates of impact point M .
 - Decompose $\vec{d}$ into $\vec{d}_{//}$ and $\vec{d}_{\perp}$.
 - Express reflected ray direction $\vec{d}_r = \vec{d}_{//} - \vec{d}_{\perp}$ in component form.
 - Verify reflection using optical reflected point S' .

4.7 Volume Problem

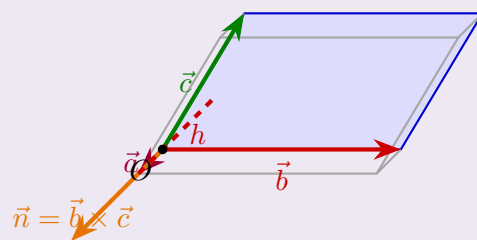
The most important geometric application of 3D vectors is calculating volume.

- The magnitude of the cross product of two base vectors gives the area of the base region.
- Projecting a third, non-coplanar vector onto a normal vector to the base yields the perpendicular height.
- Taking the scalar product of the third vector with the cross product of the two base vectors yields the volume. This operation is called the **scalar triple product**.

Theorem 4.7.1: Volume of a Parallelepiped / Prism

Let $\vec{a}$, $\vec{b}$, and $\vec{c}$ be three non-coplanar vectors representing adjacent edges meeting at a vertex of a parallelepiped. Volume V is given by the absolute value of their scalar triple product:

$$V = |\vec{a} \cdot (\vec{b} \times \vec{c})| = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix}$$



Proof. Base area $S = |\vec{b} \times \vec{c}|$. Perpendicular height $h = \frac{|\vec{a} \cdot (\vec{b} \times \vec{c})|}{|\vec{b} \times \vec{c}|}$. Volume $V = S \times h = |\vec{a} \cdot (\vec{b} \times \vec{c})|$. Expanding in component form yields the 3×3 determinant. $\square$

Example 4.7.1: Volume of a Parallelepiped

Find the volume of the parallelepiped spanned by adjacent edge vectors $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + 2\hat{k}$, and $\vec{c} = 3\hat{i} - 2\hat{j} + \hat{k}$.

Solution:

1. Evaluate $\vec{b} \times \vec{c} = 7\hat{i} + 5\hat{j} - 11\hat{k}$. 2. Scalar triple product $\vec{a} \cdot (\vec{b} \times \vec{c}) = 14 + 5 + 11 = 30$. 3. Volume $V = |30| = 30$ cubic units.

Exercise 4.7.1: Volume of a Parallelepiped

Given vectors $\vec{u} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{v} = 2\hat{i} - \hat{j} + \hat{k}$, and $\vec{w} = 3\hat{i} + \hat{j} - 2\hat{k}$.

- Evaluate scalar triple product $\vec{u} \cdot (\vec{v} \times \vec{w})$.
- Find the volume of the parallelepiped formed by $\vec{u}$, $\vec{v}$, and $\vec{w}$.

Example 4.7.2: Volume of a Triangular Prism

A triangular prism has base vectors $\vec{a} = 2\hat{i} - \hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j}$ in the xy -plane, and slant edge vector $\vec{c} = \hat{i} + \hat{j} + 3\hat{k}$. Calculate its volume.

Solution:

- Base area product $\vec{a} \times \vec{b} = 5\hat{k} \implies S = 2.5$ sq. units.
- Scalar triple product $\vec{c} \cdot (\vec{a} \times \vec{b}) = 15$.
- Volume $V = \frac{1}{2}(15) = 7.5$ cubic units.

Exercise 4.7.2: Volume of a Triangular Prism

The base of a triangular prism is formed by points $O(0, 0, 0)$, $A(2, 1, 0)$, and $B(0, 3, 0)$. Top vertex above O is $D(1, 1, 4)$.

- Find base area of $\triangle OAB$.
- Calculate volume of the triangular prism.

Example 4.7.3: Finding Height Vector Relative to a Specified Base

A parallelepiped is spanned by $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + 2\hat{k}$, and $\vec{c} = 3\hat{i} - 2\hat{j} + \hat{k}$.

- Find base area S formed by $\vec{b}$ and $\vec{c}$.
- Calculate perpendicular height h from the tip of $\vec{a}$ to base plane.
- Find height vector $\vec{h}$ perpendicular to the base plane.

Solution:

$$(a) \vec{n} = 7\hat{i} + 5\hat{j} - 11\hat{k} \implies S = \sqrt{195} \text{ sq. units.}$$

$$(b) \text{ Volume } V = 30 \implies h = \frac{30}{\sqrt{195}} = \frac{2\sqrt{195}}{13}.$$

$$(c) \vec{h} = \text{proj}_{\vec{n}} \vec{a} = \frac{30}{195}(7\hat{i} + 5\hat{j} - 11\hat{k}) = \frac{14}{13}\hat{i} + \frac{10}{13}\hat{j} - \frac{22}{13}\hat{k}.$$

Exercise 4.7.3: Finding Height Vector Relative to a Specified Base

Given a parallelepiped spanned by $\vec{u} = 3\hat{i} + 2\hat{k}$, $\vec{v} = \hat{i} + 4\hat{j}$, and $\vec{w} = 2\hat{j} + 5\hat{k}$.

- Find volume V of the parallelepiped.
- Find perpendicular height h from the tip of $\vec{u}$ to base plane spanned by $\vec{v}$ and $\vec{w}$.

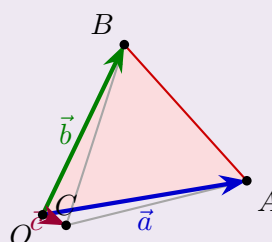
Theorem 4.7.2: Volume of Pyramids and Tetrahedra

- The volume of a **pyramid with a parallelogram base** spanned by $\vec{b}$ and $\vec{c}$ and apex vector $\vec{a}$ is:

$$V = \frac{1}{3} |\vec{a} \cdot (\vec{b} \times \vec{c})|$$

- The volume of a **triangular pyramid (tetrahedron)** with adjacent edges $\vec{a}, \vec{b}, \vec{c}$ meeting at a vertex is:

$$V = \frac{1}{6} |\vec{a} \cdot (\vec{b} \times \vec{c})|$$



Proof of (b). Using $V = \frac{1}{3} \times \text{Base Area} \times \text{Height}$ with $S = \frac{1}{2} |\vec{a} \times \vec{b}|$ yields $V = \frac{1}{6} |\vec{c} \cdot (\vec{a} \times \vec{b})|$. $\square$

Example 4.7.4: Volume of a Rectangular Pyramid

A pyramid has a parallelogram base $OABC$ on the xy -plane with $\vec{OA} = 4\hat{i}$ and $\vec{OC} = 3\hat{j}$. Apex V has position vector $\vec{OV} = 2\hat{i} + \hat{j} + 5\hat{k}$. Find its volume.

Solution:

1. $\vec{OA} \times \vec{OC} = 12\hat{k}$.
2. $\vec{OV} \cdot (\vec{OA} \times \vec{OC}) = 60$.
3. $V = \frac{1}{3}(60) = 20$ cubic units.

Exercise 4.7.4: Volume of a Rectangular Pyramid

The base of a pyramid is spanned by $\vec{u} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{v} = \hat{i} - 2\hat{j} + 3\hat{k}$. Apex vector is $\vec{w} = 3\hat{i} + 4\hat{j} + 2\hat{k}$. Calculate its volume.

Example 4.7.5: Volume of a Tetrahedron

Find the volume of tetrahedron $ABCD$ with vertices $A(1, 1, 1)$, $B(3, 2, 0)$, $C(2, 4, 1)$, and $D(4, 2, 5)$.

Solution:

1. Displacement vectors: $\vec{AB} = 2\hat{i} + \hat{j} - \hat{k}$, $\vec{AC} = \hat{i} + 3\hat{j}$, $\vec{AD} = 3\hat{i} + \hat{j} + 4\hat{k}$.
2. $\vec{AB} \times \vec{AC} = 3\hat{i} - \hat{j} + 5\hat{k}$.
3. $\vec{AD} \cdot (\vec{AB} \times \vec{AC}) = 28$.
4. Volume $V = \frac{1}{6}(28) = \frac{14}{3}$ cubic units.

Exercise 4.7.5: Volume of a Tetrahedron

Vertices of tetrahedron $PQRS$ are $P(0, 1, 2)$, $Q(2, 3, 1)$, $R(1, 0, 4)$, and $S(3, 2, 5)$.

- (a) Find displacement vectors $\vec{PQ}$, $\vec{PR}$, and $\vec{PS}$.
- (b) Calculate volume of tetrahedron $PQRS$.
- (c) Find perpendicular distance from vertex S to plane PQR .

As a final note to this section, recall that while we previously established whether three vectors are coplanar by testing for linear dependence, we now have a powerful geometric alternative: three non-zero vectors are coplanar if and only if their scalar triple product is zero.

Example 4.7.6: Coplanarity by Zero Volume

Determine whether vectors $\vec{u} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{v} = 2\hat{i} - \hat{j} + 3\hat{k}$, and $\vec{w} = 4\hat{i} + 3\hat{j} + \hat{k}$ are coplanar by evaluating their scalar triple product.

Solution:

1. $\vec{v} \times \vec{w} = -10\hat{i} + 10\hat{j} + 10\hat{k}$.
2. $\vec{u} \cdot (\vec{v} \times \vec{w}) = -10 + 20 - 10 = 0$.
3. Since the scalar triple product is zero, the spanned volume is zero; thus $\vec{u}, \vec{v}, \vec{w}$ are **coplanar**.

Exercise 4.7.6: Coplanarity by Zero Volume

Given vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} - 2\hat{k}$, and $\vec{c} = 3\hat{i} + 2\hat{j} + k\hat{k}$.

- (a) Express scalar triple product $\vec{a} \cdot (\vec{b} \times \vec{c})$ in terms of k .
- (b) Find k such that vectors $\vec{a}, \vec{b}$, and $\vec{c}$ are coplanar.

Follow-up Practice

Level 1 (Basic)

- Given vectors $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} - \hat{j} + 2\hat{k}$, and $\vec{c} = 3\hat{i} + \hat{j} - \hat{k}$.
 - Compute scalar triple product $\vec{a} \cdot (\vec{b} \times \vec{c})$.
 - Find volume of parallelepiped spanned by $\vec{a}, \vec{b}, \vec{c}$.
 - Find volume of tetrahedron spanned by $\vec{a}, \vec{b}, \vec{c}$.
- Four points in space are $O(0, 0, 0)$, $A(2, 0, 0)$, $B(1, 3, 0)$, and $C(1, 1, 4)$.
 - Show base $\triangle OAB$ lies in the xy -plane and calculate its area.
 - Calculate volume of tetrahedron $OABC$.

Level 2 (Advanced)

- Consider points $A(1, 2, 1)$, $B(3, 0, 2)$, $C(2, 4, 0)$, and $D(k, 1, 3)$, where k is a real parameter.
 - Express scalar triple product $\vec{AD} \cdot (\vec{AB} \times \vec{AC})$ in terms of k .
 - If volume of tetrahedron $ABCD$ is 5, find all possible values of k .
 - Find k such that points A, B, C , and D are coplanar.
- A tetrahedron has vertices $O(0, 0, 0)$, $A(a, 0, 0)$, $B(0, b, 0)$, and $C(0, 0, c)$ ($a, b, c > 0$).
 - Express volume V of tetrahedron $OABC$ in terms of a, b, c .
 - Find Cartesian equation of plane containing $\triangle ABC$.
 - Verify De Gua's Theorem: $[\text{Area}(\triangle ABC)]^2 = [\text{Area}(\triangle OAB)]^2 + [\text{Area}(\triangle OBC)]^2 + [\text{Area}(\triangle OCA)]^2$.

Level 3 (Challenge)

- Let L_1 and L_2 be two skew lines:

$$L_1 : \vec{r}_1(t) = (1, 2, 3) + t(2, 1, -1), \quad L_2 : \vec{r}_2(s) = (3, -1, 2) + s(1, 2, 2)$$

- Compute cross product $\vec{n} = \vec{d}_1 \times \vec{d}_2$.
- Show $|(\vec{b} - \vec{a}) \cdot (\vec{d}_1 \times \vec{d}_2)|$ measures parallelepiped volume.
- Prove shortest perpendicular distance D between skew lines is $D = \frac{|(\vec{b} - \vec{a}) \cdot (\vec{d}_1 \times \vec{d}_2)|}{|\vec{d}_1 \times \vec{d}_2|}$.
- Evaluate exact shortest distance D between L_1 and L_2 .

4.8 Comprehensive Chapter Exercises

- In tetrahedron $OABC$, let $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$, and $\vec{OC} = \vec{c}$. Points P, Q , and R divide OA, OB , and OC internally in ratios $1 : 2$, $2 : 1$, and $1 : 3$ respectively.
 - Express position vector $\vec{OG}$ of centroid G of $\triangle PQR$ in terms of $\vec{a}, \vec{b}$, and $\vec{c}$.
 - Find the ratio of the volume of tetrahedron $OPQR$ to the volume of tetrahedron $OABC$.
- Points $A(1, 0, 2)$, $B(3, 2, 0)$, and $C(0, 4, 1)$ define $\triangle ABC$ in 3D space.
 - Compute vector product $\vec{AB} \times \vec{AC}$ and deduce the area of $\triangle ABC$.
 - Find the Cartesian equation of plane Π containing $\triangle ABC$.
 - Find position vector $\vec{OH}$ of orthocenter H of $\triangle ABC$.
- Given points $A(2, -1, 3)$ and $B(4, 3, -1)$.
 - Find Cartesian equation of perpendicular bisector plane Π of segment AB .
 - Find coordinates of point K on z -axis lying on plane Π .
 - Calculate area of $\triangle ABK$.
- A straight line L is given by $\frac{x-1}{2} = \frac{y+2}{-1} = \frac{z-3}{2}$, and $P(3, 1, 4)$ is a point in space.
 - Find coordinates of foot of perpendicular N_0 from P to line L .
 - Calculate the shortest perpendicular distance from P to line L .
 - Find coordinates of reflected point P' of P across line L .
- Two skew lines are $L_1 : \vec{r} = (\hat{i} + 2\hat{k}) + t(2\hat{i} + \hat{j} - \hat{k})$ and $L_2 : \vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + s(\hat{i} - 2\hat{j} + 2\hat{k})$.
 - Find unit vector $\hat{u}$ perpendicular to both L_1 and L_2 .
 - Calculate shortest perpendicular distance between L_1 and L_2 .
 - Find points of closest approach P_1 on L_1 and P_2 on L_2 .
- Two planes are defined by $\Pi_1 : x + 2y - z = 3$ and $\Pi_2 : 2x - y + z = 4$.
 - Find acute angle θ between planes Π_1 and Π_2 .
 - Find direction vector $\vec{d}$ of line of intersection $L = \Pi_1 \cap \Pi_2$.
 - Express vector equation of line L .
- Vertices of tetrahedron $ABCD$ are $A(1, 1, 1)$, $B(3, 2, -1)$, $C(0, 4, 2)$, and $D(k, 1, 5)$.
 - Express volume $V(k)$ of tetrahedron $ABCD$ in terms of k .
 - If volume $V = 6$, find all possible values of k .
 - For $k = 2$, find perpendicular height from D to plane ABC .
- Plane Π has equation $2x - y + 2z = 5$, and vector $\vec{v} = 3\hat{i} + 4\hat{j} - \hat{k}$.
 - Decompose $\vec{v}$ into $\vec{v}_{//} + \vec{v}_{\perp}$ relative to plane Π .
 - Calculate magnitude $|\vec{v}_{//}|$.
 - Find exact angle θ between vector $\vec{v}$ and plane Π .

9. Line $L : \vec{r} = (2\hat{i} + \hat{j}) + t(\hat{i} + 2\hat{j} - 2\hat{k})$ and plane $\Pi : x + y + z = 1$.
 - (a) Find coordinates of intersection point $K = L \cap \Pi$.
 - (b) Find vector projection of direction vector of L onto plane Π .
 - (c) Write down vector equation of projected line L' on plane Π .
10. A light ray emitted from $S(1, 2, 3)$ strikes mirror plane $\Pi : x - 2y + 2z = 2$ at point $M(0, 1, 2)$.
 - (a) Find vector equation of incident ray line SM .
 - (b) Find unit direction vector of reflected ray.
 - (c) Find coordinates of reflected source point S' across plane Π .
11. Find Cartesian equation of the plane containing line $L : \frac{x-1}{2} = \frac{y+1}{3} = z-2$ and passing through point $P(2, 0, -1)$.
12. In tetrahedron $OABC$, edges OA, OB, OC are mutually perpendicular with lengths a, b, c .
 - (a) Show area of $\triangle ABC = \frac{1}{2}\sqrt{a^2b^2 + b^2c^2 + c^2a^2}$.
 - (b) Find perpendicular distance from origin O to plane ABC .
13. Vectors are given by $\vec{u} = \hat{i} + \lambda\hat{j} + \hat{k}$, $\vec{v} = \lambda\hat{i} + \hat{j} + \hat{k}$, $\vec{w} = \hat{i} + \hat{j} + \lambda\hat{k}$.
 - (a) Find all real values of λ such that $\vec{u}, \vec{v}, \vec{w}$ are coplanar.
 - (b) For $\lambda = 2$, calculate volume of spanned parallelepiped.
14. Vertices of $\triangle ABC$ are $A(2, 0, 0)$, $B(0, 4, 0)$, and $C(0, 0, 4)$.
 - (a) Find coordinates of circumcenter U of $\triangle ABC$.
 - (b) Calculate circumradius R of $\triangle ABC$.
15. A sphere has equation $S : x^2 + y^2 + z^2 - 2x + 4y - 6z = 11$, and plane $\Pi : 2x + 2y - z = 10$.
 - (a) Find center C_0 and radius R_0 of sphere S .
 - (b) Show plane Π intersects sphere S in a circle, and find its center C' and radius r' .
16. Find Cartesian equations of two angle bisector planes between non-parallel planes $\Pi_1 : 2x - y + 2z - 1 = 0$ and $\Pi_2 : x + 2y - 2z + 3 = 0$.
17. Prove that opposite edges of a regular tetrahedron are mutually orthogonal using scalar products.
18. A parallelepiped is spanned by $\vec{a} = \hat{i} + \hat{j}$, $\vec{b} = \hat{j} + \hat{k}$, and $\vec{c} = \cos\theta\hat{i} + \sin\theta\hat{j} + \hat{k}$ ($0 \leq \theta \leq 2\pi$).
 - (a) Express volume $V(\theta)$ as a function of θ .
 - (b) Find maximum volume $V_{\max}$ and corresponding values of θ .
19. Line $L : \vec{r} = (\hat{i} + 2\hat{j} - \hat{k}) + t(3\hat{i} - \hat{j} + 2\hat{k})$ and plane $\Pi : 2x + 4y + z = 7$.
 - (a) Show line L is parallel to plane Π .
 - (b) Calculate perpendicular distance between line L and plane Π .
20. Prove Lagrange's identity $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = (\vec{a} \cdot \vec{c})(\vec{b} \cdot \vec{d}) - (\vec{a} \cdot \vec{d})(\vec{b} \cdot \vec{c})$.

21. Find Cartesian equation of the sphere passing through points $O(0, 0, 0)$, $A(1, 0, 0)$, $B(0, 2, 0)$, and $C(0, 0, 3)$.
22. Let line L pass through origin along unit vector $\hat{u} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$.
 - (a) Find vector projection $\vec{p}$ of an arbitrary vector $\vec{r}$ onto line L .
 - (b) Find component vector $\vec{r}_{\perp}$ of $\vec{r}$ orthogonal to line L .
23. A pyramid $V-ABCD$ has a rectangular base. Plane Π intersects lateral edges VA, VB, VC, VD at A', B', C', D' such that $\frac{VA'}{VA} = k_1, \frac{VB'}{VB} = k_2, \frac{VC'}{VC} = k_3, \frac{VD'}{VD} = k_4$. Show that $\frac{1}{k_1} + \frac{1}{k_3} = \frac{1}{k_2} + \frac{1}{k_4}$.
24. An isosceles tetrahedron $ABCD$ satisfies $AB = CD = a, AC = BD = b, AD = BC = c$. Prove that all four faces are congruent acute triangles, and express its volume in terms of a, b, c .
25. A moving particle P has position vector $\vec{r}(t) = (t+1)\hat{i} + (2t-1)\hat{j} + t^2\hat{k}$ at time $t \geq 0$.
 - (a) Find velocity vector $\vec{v}(t)$.
 - (b) Find minimum distance from particle P to plane $\Pi : x + 2y - z = 0$.
26. Three planes are $\Pi_1 : x + y + z = 1, \Pi_2 : 2x + 3y - z = 2, \Pi_3 : x + 2y - 2z = k$.
 - (a) Find k such that the three planes intersect along a common line L .
 - (b) Write down vector equation of line L .
27. Given skew lines $L_1 : \vec{r}_1 = t(\hat{i} + \hat{j})$ and $L_2 : \vec{r}_2 = \hat{k} + s(\hat{j} + \hat{k})$. Find parametric equations of common perpendicular line segment P_1P_2 connecting L_1 and L_2 .
28. For $\triangle ABC$ in 3D space with projection triangle areas S_{xy}, S_{yz}, S_{xz} on coordinate planes, prove $S^2 = S_{xy}^2 + S_{yz}^2 + S_{xz}^2$.
29. A force $\vec{F} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ is applied at $P(3, -1, 2)$. Calculate moment of force vector $\vec{r}$ about origin O .
30. In tetrahedron $OABC$, $\vec{OA} = \vec{a}, \vec{OB} = \vec{b}, \vec{OC} = \vec{c}$. Point M is midpoint of AB , N lies on OC with $ON : NC = 1 : 2$. Plane Π passes through O, M, N .
 - (a) Find a normal vector to plane Π in terms of $\vec{a}, \vec{b}, \vec{c}$.
 - (b) Find the ratio in which plane Π divides edge BC .
 - (c) Calculate the ratio of volumes of the two solid regions into which plane Π divides tetrahedron $OABC$.